Do Markets Discipline the Adoption of Dangerous Technology?*

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Abstract

We study technology adoption when disaster risk rises with the technology's capability and deployment. A survivable disaster that imposes the same proportional loss on firms that adopt the new technology and those that do not leaves relative payoffs unchanged while raising marginal utility. Disasters are more likely in high-capability states, so they tilt valuation toward states in which adopting firms are relatively more productive. This raises their values and encourages adoption, which further raises disaster risk. Equilibrium deployment therefore exceeds the level that would prevail if the technology posed no disaster risk. Disaster losses confined to adopting firms give them lower payouts in high-marginal-utility states and can significantly discipline adoption. An extinction-level disaster eliminates all financial claims, but can still encourage adoption under recursive preferences. The potential loss of future consumption can raise current marginal utility in high-capability states, again increasing the value of adopting firms. Empirically, option-implied tail risk now moves with stock prices in a manner broadly consistent with a potential AI disaster.

Keywords: artificial intelligence, technology adoption, disaster risk, state prices, option-implied tails.
JEL codes: G12, O33, D62.

1 Introduction

The adoption of new technologies can increase productivity while also creating new dangers for society as a whole. Individual firms may therefore decide to adopt a new technology even when this adoption is not socially optimal, because they do not fully internalize the dangers their adoption creates (Acemoglu and Lensman, 2024; Jones, 2024). We show that financial markets can amplify this incentive rather than restrain it. In particular, if AI's productivity and its potential for causing a disaster are both increasing in its capability, then disaster states are more likely to be those where AI-adopting firms are relatively more productive. The high marginal utility of these states increases the value of adoption, potentially leading to more deployment than if AI posed no disaster risk at all.

This pricing logic parallels results in the climate-finance literature. Which technology performs relatively well in bad states determines whether market pricing reinforces or offsets the externality. Baker et al. (2022) establish that hedging demand can encourage investment in the activity

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generating the harm: more productive polluting firms generate both higher payoffs and more environmental damage, making their stocks a hedge that attracts more capital. We argue that this incentive can also operate for AI adoption. When an economy-wide survivable disaster imposes the same proportional loss on adopters and non-adopters, it reduces dividends and aggregate consumption without changing relative payoffs. Investors pay a premium for firms that perform relatively well in the states where calamity is most likely, even when those firms' deployment increases the danger. If damage instead falls only on adopters, their direct exposure works against this incentive and substantially disciplines adoption in our numerical exercises. This case echoes Pástor et al. (2021), where worsening climate conditions can bring regulation that penalizes brown firms, giving green firms a hedging advantage and cheaper capital. Here, non-adopters play the green firms' role, so hedging demand discourages adoption.

The comparison becomes more subtle when a disaster eliminates all financial claims. Following Chow et al. (2026), we also study this extinction case. Under CRRA preferences, investors' willingness to pay to avoid extinction does not affect private prices, because no financial claim pays after the event. With recursive preferences, however, the lost value of future consumption enters current marginal utility. The prospect of extinction can therefore create a hedging incentive similar to that under an economy-wide survivable disaster, encouraging adoption even though all financial claims disappear.

We first establish the mechanism in a static, fixed-date adoption model built on Pástor and Veronesi (2009), with individual firm adoption as in Gutierrez and Ready (2026). The simple setting, with CRRA preferences, lets us show analytically that the risk of a disaster imposing the same proportional loss on all firms encourages private adoption. When potential losses fall only on adopters, the theoretical sign is ambiguous, but adoption falls substantially in the numerical illustration. Extinction risk instead reduces the probability of receiving high-capability payouts and therefore discourages adoption. This effect is modest because it operates through physical survival probabilities: no financial claim pays in extinction, so there is no marginal-utility amplification of those missing payouts.

We then develop an illustrative quantitative exercise with endogenous adoption, learning, intermediate consumption, and recursive preferences. Its purpose is to provide some context for the potential economic size of these effects. Economy-wide survivable losses significantly accelerate adoption and adopter-only damage substantially restrains it, as in the fixed-date illustration. Recursive preferences change how the valuation of extinction enters financial prices. The welfare cost of catastrophic loss of life can be large when assessed using estimates of the value of a statistical life (Martin and Pindyck, 2021). Yet under CRRA, even a large willingness to pay to prevent extinction leaves private prices unchanged at a given allocation. No financial claim pays after extinction, and marginal utility in surviving states is unchanged. Recursive preferences break this separation. The adoption response reflects two competing pricing effects. News about the loss of future consumption can raise the pricing weight on high-capability states, increasing the value of adopters as a hedge. At the same time, extinction makes future financial payments less likely and can reduce their value

relative to consumption today. This discounting effect works against adoption, which requires a current conversion cost in exchange for future dividends.

We consider two specifications, distinguished by how willingness to pay for avoiding extinction changes with prosperity. In the first, extinction ends the consumption problem, following the no-bequest mortality specification of Li and Smetters (2010). Holding consumption growth and disaster probabilities fixed, investors would surrender the same fraction of consumption to eliminate extinction risk as the economy becomes richer. The absolute cost grows with prosperity, but its proportional cost does not. In our quantitative exercise, the discounting effect remains modest enough for the hedging incentive to bring adoption forward.

Alternatively, we assign extinction the utility of a fixed nonfinancial consumption stream. One interpretation is that people survive at subsistence, but the modern financial system disappears.[1] A wealthier society would surrender a larger fraction of consumption to avoid falling to that same fixed living standard.[2] All financial claims provide nothing in that outcome and pay only if the economy survives. As the gap between surviving prosperity and the fixed fallback widens, investors require a higher return to give up consumption today for those future payments. In our illustration, this stronger discounting outweighs the hedging incentive and generally reduces deployment relative to the no-disaster benchmark.

Finally, we examine whether favorable technology-stock returns coincide with increases in both upside and downside option-implied tail probabilities. The economy-wide survivable case predicts increases in both tails, with a stronger downside asymmetry for firms that have not yet adopted. The recent co-movement of technology-stock returns and index tail probabilities is broadly consistent with this pattern. However, we acknowledge that many forces could generate such co-movement, so we view this evidence as suggestive.

Our analysis connects the asset pricing of technological revolutions to the deployment externality studied by Acemoglu and Lensman (2024). Gutierrez and Ready (2026) show that when adoption changes only systematic risk, the private and planner conditions coincide because that risk is capitalized in firm values. We introduce a deployment externality and examine how pricing can increase the private return to the risky action. This incentive complements the strategic forces studied in recent work on AI development. Bueno de Mesquita et al. (2026) analyze firms' allocation of resources between speed and safety, while Fudenberg and Koh (2026b,a) study how transparency and strategic incentives shape racing and restraint. Our pricing incentive operates without strategic rivalry. Financial markets can weaken disaster mitigation through other channels as well: Baker et al. (2020) show how disagreement and private insurance reduce support for disaster-prevention policy.

[1] For a similar approach, see the random-horizon model of Aurand and Huang (2023).

[2] Another interpretation of the two specifications concerns the income elasticity of willingness to pay for survival: unit elasticity in the first case, where extinction ends the consumption problem, and above one in the second, where people survive on a fixed nonfinancial consumption stream. This connects to the value of a statistical life (VSL), defined as marginal willingness to pay for a reduction in mortality risk. Martin and Pindyck (2021) discuss microeconomic estimates of VSL income elasticities around 0.5–0.6, whereas Hall and Jones (2007) and Jones (2016) provide mechanisms through which the value of survival rises faster than consumption.

Related models connect innovation to displacement risk and asset returns (Gârleanu et al., 2012; Kogan et al., 2020). Chen (2026) studies AI stocks as hedges against consumption displacement and shows how incomplete insurance can lead households to block socially beneficial AI development. In our setting, hedging demand instead encourages deployment that raises disaster risk, even though disasters also harm adopters.

A growing literature uses asset prices and investment to assess expectations about AI (Eisfeldt et al., 2023; Andrews and Farboodi, 2025; Wachter and Wachter, 2026). Related work explains how AI can lower yields through stronger fiscal backing of government debt (Kung et al., 2026) or labor displacement with limited capital-market participation (Maresca, 2026). Shen (2026) examines the pricing of AI-risk news factors. Our question concerns how the pricing of AI-linked danger feeds back into technology deployment. Our empirical analysis examines the co-movement of technology-stock returns and option-implied tail probabilities.

Section 2 presents the static model, incidence comparisons, and risk-neutral distributions. Section 3 develops the dynamic model and simulations. Section 4 examines technology stocks and tail risks in the data.

2 Simple Model

We begin with an illustrative model in which firms can adopt a new technology at a fixed date. The technology and learning environment follows Pástor and Veronesi (2009), with individual firm adoption as in Gutierrez and Ready (2026). Firms differ in their adoption costs, so adoption need not occur throughout the economy at once. Fixing the adoption date and retaining CRRA preferences lets us isolate analytically how disaster risk changes the incentive to adopt. Section 3 allows firms to choose when to adopt and separates risk aversion from intertemporal substitution using recursive preferences. That extension also requires a more explicit treatment of utility after extinction.

2.1 Environment and adoption

A representative household owns a unit mass of firms. Firms reinvest their output until the terminal date T, when their accumulated capital is paid to the household and consumed. The household has utility

$$U(C_T) = \frac{C_T^{1-\gamma}}{1-\gamma}, \qquad \gamma > 1,$$

where C_T is aggregate terminal consumption and γ measures risk aversion.

At the review date $t^* < T$, each firm chooses whether to retain the old technology or irreversibly adopt the new one. We first describe payoffs without technological disasters. We condition on current capital and productivity and express terminal payoffs relative to the common old-technology payoff, normalized to one. Following the fixed-date adoption setup of Pástor and Veronesi (2009), adoption changes long-run productivity but leaves exposure to the old-economy productivity shock

unchanged. We suppress this shock because it affects adopting and non-adopting firms proportionally and factors out of the private adoption conditions in all three disaster cases.[3] Investors continue to learn about capability from new-technology productivity relative to the old-technology benchmark. We therefore focus on uncertainty about capability and the technological disaster risk introduced below.

The new technology's capability is measured by ψ, its effect on long-run productivity. Higher capability makes adoption more productive. Before adoption costs, switching technologies multiplies a firm's terminal payoff by

$$e^{A_2(\Delta)\psi}, \qquad \Delta = T - t^*.$$

Here Δ is the time remaining after adoption, and $A_2(\Delta)$ measures how much of the productivity gain accumulates over that horizon. Following Pástor and Veronesi (2009),

$$A_2(\Delta) = \Delta - \frac{1 - e^{-\phi\Delta}}{\phi}, \tag{1}$$

where $\phi > 0$ is the speed at which productivity adjusts toward its long-run mean.

Capability is unknown when firms adopt. Observations from a small pilot sector give all firms and the household the same beliefs,

$$\psi \mid \mathcal{F}_{t^*} \sim \mathcal{N}(\widehat{\psi}, \widehat{s}^2).$$

The information available at adoption is $\mathcal{F}_{t^*}$. The posterior mean $\widehat{\psi} = \widehat{\psi}_{t^*}$ measures expected capability and responds to observed productivity news. The posterior variance $\widehat{s}_t^2$ falls deterministically as information accumulates, so its value at adoption, $\widehat{s}^2 = \widehat{s}_{t^*}^2$, is pinned down by the exogenous review date and the learning parameters. At that date, firms' adoption decisions depend on the realized posterior mean $\widehat{\psi}$. Throughout this section, expectations use these review-date beliefs. The pilot sector does not contribute materially to aggregate wealth. Appendix B provides the underlying productivity and learning processes.

Firms differ in the cost of adopting. Index firms by u, uniformly distributed on $[0, 1]$, and let

$$\kappa_\epsilon(u) = \kappa + \epsilon(2u - 1) \tag{2}$$

be the fraction of capital lost when firm u converts. The average conversion cost is κ, and costs range from $\kappa - \epsilon$ to $\kappa + \epsilon$, with $0 < \kappa - \epsilon < \kappa + \epsilon < 1$. A firm's terminal payoff, measured relative

[3] Let $B > 0$ denote the common old-technology terminal payoff, independent of capability and disaster arrivals conditional on review-date information, and suppose $\mathbb{E}[B^{1-\gamma}]$ is finite. With CRRA utility, multiplying consumption and each firm's payoff by B multiplies every private marginal adoption gain by the same positive factor $\mathbb{E}[B^{1-\gamma}]$, leaving its zero unchanged. This factorization also preserves planner adoption under survivable disasters. Extinction welfare additionally depends on whether reservation consumption is fixed in levels or scales with B. Absolute valuations and payoff distributions need not be unchanged.

to what it would receive without adopting, is therefore

$$\begin{cases} 1, & \text{if it retains the old technology,} \\ [1-\kappa_\epsilon(u)]e^{A_2(\Delta)\psi}, & \text{if it adopts.} \end{cases}$$

Low-cost firms adopt first. If a fraction q adopts, firms with $u \leq q$ use the new technology and the marginal adopter has type $u = q$. Summing firm payoffs gives aggregate terminal consumption in the absence of disasters,

$$H(q,\psi) = \underbrace{1-q}_{\text{non-adopters}} + \underbrace{\int_0^q [1-\kappa_\epsilon(u)]e^{A_2(\Delta)\psi}\,du}_{\text{adopters}}. \tag{3}$$

Thus H includes both the productivity gains from adoption and the capital lost in conversion.

The additional terminal payoff from converting the marginal firm is

$$H_q(q,\psi) = [1-\kappa_\epsilon(q)]e^{A_2(\Delta)\psi} - 1. \tag{4}$$

The notation H_q denotes the derivative of aggregate consumption with respect to the adopted fraction. Economically, it is the adopter's payoff minus the old-technology payoff it gives up.

Firms choose adoption to maximize market value. Because the household consumes aggregate firm payoffs, terminal payments are valued using marginal utility $H(q,\psi)^{-\gamma}$. At an interior equilibrium, the marginal firm is indifferent when

$$\mathbb{E}\left[\underbrace{H(q,\psi)^{-\gamma}}_{\text{marginal utility}} \underbrace{H_q(q,\psi)}_{\text{payoff gain from adoption}}\right] = 0. \tag{5}$$

For a given adopted fraction q, this condition defines the belief threshold at which the marginal firm adopts.

The planner faces the same condition. An additional adoption changes aggregate consumption by H_q, which the planner values at marginal utility $H^{-\gamma}$. Firms therefore internalize the effect of adoption on ordinary consumption risk through market prices. Introducing disaster risk that depends on aggregate deployment breaks this equivalence.

2.2 Economy-wide disaster risk and private adoption

We now add our central departure. Let N denote the number of technological disasters between t^* and T. Let $\lambda(\psi) > 0$ denote the annual disaster intensity at full deployment, with $\lambda'(\psi) > 0$, and let $\Lambda(q,\psi)$ denote the integrated intensity conditional on deployment and capability. After the

adoption decision,

$$N \mid (q,\psi) \sim \text{Poisson}(\Lambda(q,\psi)),$$
$$\Lambda(q,\psi) = \underbrace{q}_{\text{deployment}} \underbrace{\lambda(\psi)}_{\text{capability}} \Delta, \qquad \lambda'(\psi) > 0. \tag{6}$$

Both capability and use increase danger. The first makes a system more able to cause harm; the second puts more activity at risk. We use linear deployment exposure, so the hazard is zero at zero adoption but its marginal increase is positive. Risk then rises as adoption spreads. An individual firm takes q and hence $\Lambda(q,\psi)$ as given when it adopts. The planner instead recognizes that increasing q raises the economy-wide hazard. This is the new externality.

We consider three disaster structures: a economy-wide survivable disaster, an adopter-only survivable disaster, and extinction. The three cases differ in which cash flows survive the event.

Assume the valuation expectations below are finite. Suppose every disaster multiplies every firm's terminal payoff by the same survival factor $\omega \in (0,1)$. Let the superscript C label this economy-wide-disaster economy. Aggregate terminal consumption is

$$C_T^C = H(q,\psi)\omega^N. \tag{7}$$

Adopters are not physically safer than non-adopters. Let χ measure how a single disaster changes marginal-utility-weighted consumption, and define $R_D(q,\psi)$ as the expected state-price multiplier generated by the Poisson disaster process:

$$\chi = \omega^{1-\gamma} - 1 > 0, \qquad R_D(q,\psi) \equiv \mathbb{E}\Big[\omega^{(1-\gamma)N} \mid q,\psi\Big] = \exp\{\Lambda(q,\psi)\chi\}. \tag{8}$$

The multiplier combines a lower physical payoff with higher marginal utility. With $\gamma > 1$, the marginal-utility increase dominates, so $R_D > 1$. Higher capability increases this weight by making disasters more likely. This does not mean every high-productivity state has low consumption: it means economy-wide disasters shift valuation, relative to the no-disaster comparison, toward the high-capability realizations in which adopters do relatively well. After integrating over the number of disasters, write G_D^C for the decentralized marginal adoption gain; the subscript D denotes the decentralized allocation and P below denotes the planner. The adoption condition becomes

$$G_D^C(q,\widehat{\psi}) = \mathbb{E}\big[R_D(q,\psi)H(q,\psi)^{-\gamma}H_q(q,\psi)\big] = 0. \tag{9}$$

The loss cancels from the adopter–non-adopter payoff ratio, but a low-consumption payment still receives more weight in the market-value comparison. A disaster is a high-marginal-utility state. Because $\lambda'(\psi) > 0$, disaster states disproportionately select high-ψ realizations, precisely the states in which adoption is most productive.

Proposition 1 (Economy-wide disaster risk lowers the private threshold, more so when risk is greater)**.** *Fix the deployed fraction of firms at $q > 0$. Technology quality is ψ, and its posterior mean $\widehat{\psi}$ summarizes current beliefs. Let $\lambda(\psi)$ be disaster intensity at full deployment and $\omega \in (0,1)$ the*

fraction of each firm's payoff retained after a economy-wide disaster. Suppose risk aversion exceeds one ($\gamma > 1$), intensity rises with quality ($\lambda'(\psi) > 0$), posterior uncertainty is nondegenerate, and valuation expectations are finite. In each economy compared, assume the marginal private adoption gain crosses zero once from below as $\widehat{\psi}$ rises.

The following conclusions hold:

(i) The belief threshold for private adoption is strictly lower with economy-wide survivable disaster risk than without it.

(ii) The threshold falls further when the economy-wide loss becomes more severe (lower ω) or intensity increases proportionally, holding other primitives fixed.

(iii) At zero deployment, $q = 0$, these fixed-date thresholds coincide.

Proof sketch: the covariance effect. At fixed deployment $q > 0$, recall three payoff objects: H is aggregate wealth without a disaster, $H_q = \partial H/\partial q$ is the payoff gain from marginal adoption, and R_D is the disaster-induced multiplier on valuation weights. Risk aversion is γ, so the no-disaster marginal-utility weight is $H^{-\gamma}$. Evaluate these objects at the no-disaster adoption threshold. Let P be the posterior probability measure and define its marginal-utility-weighted counterpart Q_0 by

$$\frac{dQ_0}{dP} = \frac{H^{-\gamma}}{\mathbb{E}[H^{-\gamma}]}, \qquad \mathbb{E}^{Q_0}[H_q] = 0,$$

Here $\mathbb{E}^{Q_0}$ denotes expectation under Q_0. The private marginal gain under economy-wide disasters, G_D^C, then satisfies

$$\frac{G_D^C}{\mathbb{E}[H^{-\gamma}]} = \mathbb{E}^{Q_0}[R_D H_q] = \mathrm{Cov}^{Q_0}(R_D, H_q) > 0. \tag{10}$$

The covariance is computed under Q_0. Both the disaster valuation multiplier R_D and the marginal adoption payoff H_q increase strictly with capability ψ. Disaster risk therefore places more valuation weight on the realizations in which adoption pays most. The firm prefers adoption at the old threshold, so single crossing implies a lower threshold. Greater severity or a proportional intensity increase adds another increasing tilt to the valuation weights; the same covariance argument gives the further threshold reduction. Appendix A provides the full proof and comparative statics.

Remark (capability versus deployment). If intensity is capability independent, write its constant full-deployment value as $\bar{\lambda}$. Then $R_D(q, \psi) = \exp\{q\bar{\lambda}\Delta\chi\}$ factors out of the private condition, leaving the no-disaster threshold unchanged. A capability-independent additive increase in intensity likewise cancels. The planner wedge remains because deployment raises the hazard. The private threshold result requires capability-dependent risk.

The restriction $\lambda'(\psi) > 0$ captures the idea that more capable systems are both more useful and able to cause harm on a larger scale. Capability may instead arrive together with better monitoring, alignment, or control. If those improvements are strong enough to make disaster risk independent

of capability, the private hedging channel disappears; if they make risk decline with capability, the channel reverses. The deployment externality remains whenever aggregate use raises the hazard.

2.3 Incidence and the planner

The planner sees the same state-price selection effect but also internalizes the dependence of disaster intensity on aggregate deployment. Recall that H is aggregate terminal wealth, H_q is the marginal conversion payoff, and R_D is the disaster valuation multiplier. Write V_P^C for the planner's expected utility in the economy-wide-loss economy:

$$V_P^C(q) = \frac{1}{1-\gamma}\mathbb{E}\big[H(q,\psi)^{1-\gamma}R_D(q,\psi)\big]\,. \tag{11}$$

With $\chi = \omega^{1-\gamma} - 1$ as defined above, the planner marginal gain $G_P^C = \partial V_P^C/\partial q$ satisfies

$$G_P^C(q,\widehat{\psi}) = \mathbb{E}\left[R_D H^{-\gamma}\left(H_q - H\frac{\lambda(\psi)\Delta\chi}{\gamma-1}\right)\right] = 0. \tag{12}$$

The first term is exactly the decentralized condition. The second is the marginal social cost of the additional disaster hazard created by deployment. Hence the same disaster risk that makes private adoption more attractive makes the planner more reluctant to deploy.

This deployment externality is the planner mechanism of Acemoglu and Lensman (2024). Proposition 1 adds the state-price selection result.

Proposition 2 (The planner always values marginal adoption less)**.** *Let G_P^C and G_D^C be the planner and private marginal gains under a economy-wide loss. For every adopted share $q > 0$ with $\lambda(\psi) > 0$, $\omega \in (0,1)$, and $\gamma > 1$,*

$$G_P^C(q,\widehat{\psi}) < G_D^C(q,\widehat{\psi}). \tag{13}$$

At any interior decentralized allocation satisfying $G_D^C = 0$, the planner strictly prefers less adoption. If each marginal condition crosses zero once from above as q rises, then $q_P < q_D$, where q_P and q_D are planner and decentralized adoption, respectively.

Proofs are in Appendix A.

Adopter-only losses. Now put the loss on adopters alone. Their productivity advantage is no longer protected in relative terms: a disaster damages the adopted claim while leaving the old claim intact. State-price selection still favors capable technology, but it must overcome this direct loss. A lower private threshold is therefore possible, not guaranteed.

Suppose a survivable disaster damages only capital operated with AI. We use the superscript A for this adopter-only disaster economy. Recall that N counts disasters and ω is the fraction of affected payoffs retained per event. To write adopted payoffs compactly, define total adopted capital after conversion costs as

$$K(q) = \int_0^q [1 - \kappa_\epsilon(u)]\,du. \tag{14}$$

Conditional on $N = n$, let C_n^A denote aggregate consumption and $H_{q,n}^A$ the payoff gain from marginal adoption:

$$C_n^A = 1 - q + K(q)e^{A_2(\Delta)\psi}\omega^n, \tag{15}$$

$$H_{q,n}^A = [1 - \kappa_\epsilon(q)]e^{A_2(\Delta)\psi}\omega^n - 1. \tag{16}$$

Let G_D^A denote the private marginal gain in this adopter-only economy. Its adoption condition is

$$G_D^A(q, \widehat{\psi}) = \mathbb{E}_{\psi,N}\big[(C_N^A)^{-\gamma} H_{q,N}^A\big] = 0. \tag{17}$$

Disaster selection still raises the state-price weight on high-ψ states, but the adopting firm now suffers direct damage relative to a non-adopter. Let Q_0 denote the no-disaster measure with density proportional to $H^{-\gamma}$, and let Q_A be the joint measure over (ψ, N) proportional to $f_P(\psi)\Pr(N = n \mid q, \psi)(C_n^A)^{-\gamma}$, where f_P is the review-date posterior density of quality. At the no-disaster threshold, adopter-only disaster risk encourages private adoption if and only if

$$\mathbb{E}^{Q_A}[e^{A_2(\Delta)\psi}\omega^N] > \mathbb{E}^{Q_0}[e^{A_2(\Delta)\psi}]. \tag{18}$$

The sign is therefore ambiguous: the disaster-state selection gain must exceed the direct physical damage to adopters.

The planner also accounts for the extra damage that deployment imposes on existing adopters. If an additional event lowers consumption, this term makes the planner value marginal adoption less than the firm. The derivative is given in Appendix C.

Extinction. Extinction changes which payouts exist, rather than their relative size after a survivable event. Both technologies pay zero after termination, so a firm can earn its technology advantage only in surviving states. If survival is less likely at high capability, precisely those valuable adopter payouts are less likely to be received.

Use the superscript E for the extinction economy. The financial-claim mechanism follows the extinction logic in Chow et al. (2026): extinction ends consumption and the payoffs of long-lived financial claims. Let the indicator S equal one if civilization survives and zero if extinction occurs, and define the survival probability $s(q, \psi)$ by

$$s(q, \psi) = \Pr(S = 1 \mid q, \psi) = e^{-\Lambda(q,\psi)}. \tag{19}$$

Recall that H is surviving aggregate wealth and H_q is the marginal adoption payoff. Since both alternatives pay zero after extinction, the private marginal gain G_D^E weights that payoff by survival probability:

$$G_D^E(q, \widehat{\psi}) = \mathbb{E}\big[s(q, \psi)H(q, \psi)^{-\gamma} H_q(q, \psi)\big] = 0. \tag{20}$$

Corollary 1 (Extinction raises the private threshold under CRRA)**.** *In the fixed-date CRRA model,*

let risk aversion be any $\gamma > 0$ and fix deployment at $q > 0$. Suppose disaster intensity increases strictly with capability, posterior uncertainty is nondegenerate, and valuation expectations are finite. If the private marginal adoption gain in each economy crosses zero once from below as the posterior mean rises, the adoption threshold is strictly higher with extinction risk than in the no-disaster benchmark. At $q = 0$, the thresholds coincide.

Proof. At the no-disaster threshold, use the measure Q_0 from Proposition 1, whose density is proportional to $H^{-\gamma}$ and under which $\mathbb{E}^{Q_0}[H_q] = 0$. The extinction marginal gain satisfies

$$\frac{G_D^E}{\mathbb{E}[H^{-\gamma}]} = \mathbb{E}^{Q_0}[sH_q] = \mathrm{Cov}^{Q_0}(s, H_q) < 0.$$

Survival probability s decreases strictly with capability, while the marginal adoption payoff H_q increases strictly. The firm therefore prefers not to adopt at the no-disaster threshold; single crossing implies a higher threshold. At zero deployment, $s = 1$ and the conditions coincide. Unlike the economy-wide survivable result, this argument does not require $\gamma > 1$: extinction removes payoffs through physical survival probabilities, whereas the economy-wide survivable channel requires the marginal-utility increase to outweigh the proportional payoff loss.

Under the stated conditions, economy-wide survivable risk lowers the private threshold, extinction raises it, and adopter-only damage has an ambiguous effect governed by (18).

Under CRRA, risk aversion determines how we value differences in consumption, but does not determine how we value continued existence. For welfare, we therefore assign a separate reservation-consumption level $\underline{C} > 0$ after extinction. It keeps CRRA utility finite and is not corporate property; all financial claims still pay zero. The planner accounts for the loss of survival probability caused by deployment, a negative marginal effect when surviving utility exceeds reservation utility. Lowering the reservation level makes the planner more reluctant to deploy without changing private prices at a given allocation. It can make the welfare penalty arbitrarily large; with linear deployment exposure, a finite penalty can make zero adoption optimal. Appendix C gives the payoff and welfare expressions, and Appendix F explains the separate role of reservation utility.

Extinction works through the survival probability, rather than an adopter payoff multiplier. It is not the $\omega \to 0$ limit of a survivable disaster with a claim that continues to pay.

Figure 1 summarizes the central result by plotting deployment against the posterior mean of capability. At a given belief, economy-wide survivable losses raise market adoption above the no-disaster benchmark, while the planner chooses less deployment because it internalizes the additional hazard. The planner begins adopting at a capability belief of about 1.42%, well above the market entry threshold. Adopter-only damage instead lowers market adoption in this calibration. Extinction lowers it more modestly, although the planner chooses no deployment over the displayed belief range under the reservation-utility convention used here. The figure uses mean conversion cost $\kappa = 0.10$ and linear deployment exposure, as in Section 3, and fixed 20% survivable losses; Section 3 allows random loss sizes. Appendix D derives the corrective deployment charge that

Table 1: Shared calibration and model-specific assumptions

Parameter	Fixed-date exercises	Dynamic exercises
Risk aversion	4	4
Conversion costs	8.5–11.5%	8.5–11.5%
Horizon; comparison date	20; year 4	30; year 4
Prior SD; signal volatility	0.014204; 0.03	0.014204; 0.03
Old-economy volatility	0.02	0.02
Hazard; scale	$q\lambda(\psi)$; 2.25	$q\lambda(\psi)$; 2.25
Survivable loss	Fixed 20%	Random; mean 22.3%
Consumption and preferences	Terminal; CRRA	Intermediate; recursive, IES 2

Both fixed-date figures use the same calibration, including $\Delta = 16$, $A_2 = 13.1935$, and posterior SD 0.01346. Capability is measured as the long-run growth advantage in both models. Static productivity adjusts at speed $\phi = 0.3551$, while dynamic adopters receive that growth advantage immediately. The signal processes imply year-four posterior standard deviations of 0.01346 and 0.01031, respectively, before disaster information. In both models, reservation consumption is fixed at 60% of initial consumption.

closes the market–planner gap.

Disaster states are rare at the reference deployment level, helping explain the modest private extinction effect. Conditional on capability, extinction enters through the physical survival probability $e^{-\Lambda(q,\psi)}$. No financial claim pays in extinction, so there is no marginal-utility amplification of extinction-state payouts. Economy-wide survivable losses, by contrast, receive the high state-price weight associated with low consumption. Surviving payoffs still carry $H^{-\gamma}$; it is the survival factor that is physical rather than risk-neutral. Small private effects can coexist with large welfare costs.

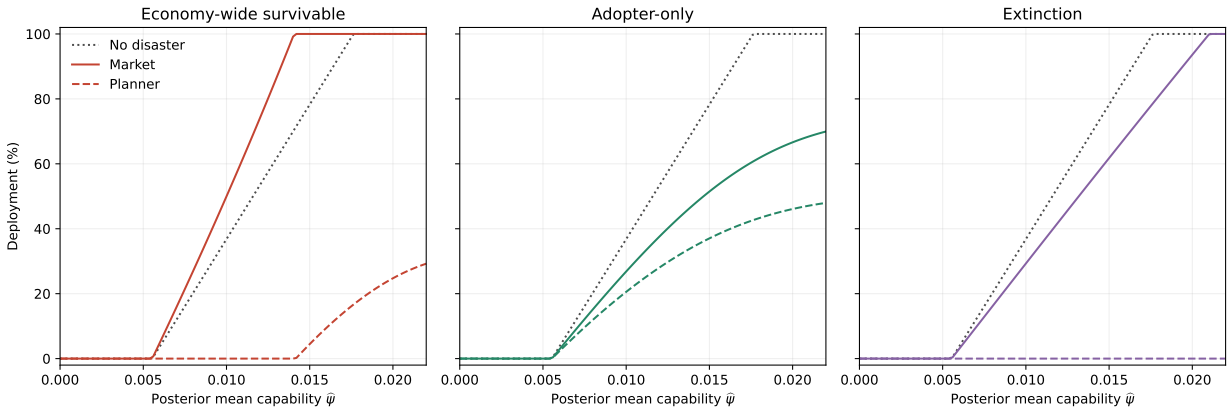


Figure 1: **Fixed-date deployment as a function of beliefs.** The panels show economy-wide survivable losses, adopter-only losses, and extinction. Solid curves show market deployment, dashed curves show planner deployment, and the dotted curve shows the no-disaster market benchmark. Parameters are given in Table 1.

2.4 Risk-neutral payoff distributions

The numerical exercises share the parameters in Table 1. In the fixed-date model we keep the survivable loss fixed at 20% to isolate the pricing mechanism. Section 3 relaxes this assumption by allowing random disaster sizes, a more realistic description of uncertain damage, along with intermediate consumption and endogenous adoption timing.

The adoption result concerns the relative value of two technologies. Options concern the whole distribution of a claim's payoff. We connect the two by comparing a low-cost firm, a high-cost firm, and an aggregate consumption claim at two review-date beliefs. Deployment is zero at the lower belief and 50% at the higher belief. The low-cost firm adopts only in the higher state; the high-cost firm remains a non-adopter in both. The adoption decisions are then held fixed while subsequent payoffs are priced.

Figure 2 shows risk-neutral densities, meaning physical probabilities weighted by the household's marginal utility and normalized. Denote this normalized risk-neutral measure by Q; every claim uses it, and $\mathbb{E}^Q$ denotes expectation under it. A low-consumption outcome receives more weight, even if its physical probability is small. The illustration restores continuous old-economy risk, which cancels from the adoption comparison but makes non-adopter payoffs continuous as well. Appendix G gives the payoff equations and pricing weights.

Both tails broaden for all three claims as deployment rises. The adopter's distribution broadens on both sides because adoption exposes its payoffs to uncertainty about technology productivity as well as disaster losses. Its expansion is approximately symmetric: high capability raises productivity while also making disasters more likely, so the productivity gain partly offsets the loss in disaster states. The non-adopter receives no such gain, and its disaster exposure produces a more pronounced downside expansion. At 50% adoption, the aggregate claim combines these two exposures and displays an intermediate pattern.

The lower row separates the higher-belief density into probability-weighted no-disaster and disaster components, which sum to the red curve above. Disaster outcomes are shifted toward lower payoffs, particularly for the non-adopter. They also receive greater weight under risk-neutral pricing: the physical probability of at least one disaster is 12.18%, compared with a risk-neutral probability of 22.25%, because marginal utility is high when consumption is low.

3 Dynamic Adoption and Recursive Preferences

The fixed-date model isolates the pricing incentive. We now allow firms to wait, learn, and adopt at different dates, with consumption paid throughout. We extend the horizon from 20 to 30 years. Intermediate consumption allows households to benefit from technology gains throughout this longer period, whereas the fixed-date model pays consumption only at the terminal date. This is an illustrative quantitative exercise aimed at gauging the size of the deployment effects, rather than a fully calibrated assessment of AI disaster risk. We use recursive preferences (Epstein and Zin, 1989) to distinguish risk aversion from the willingness to substitute consumption across

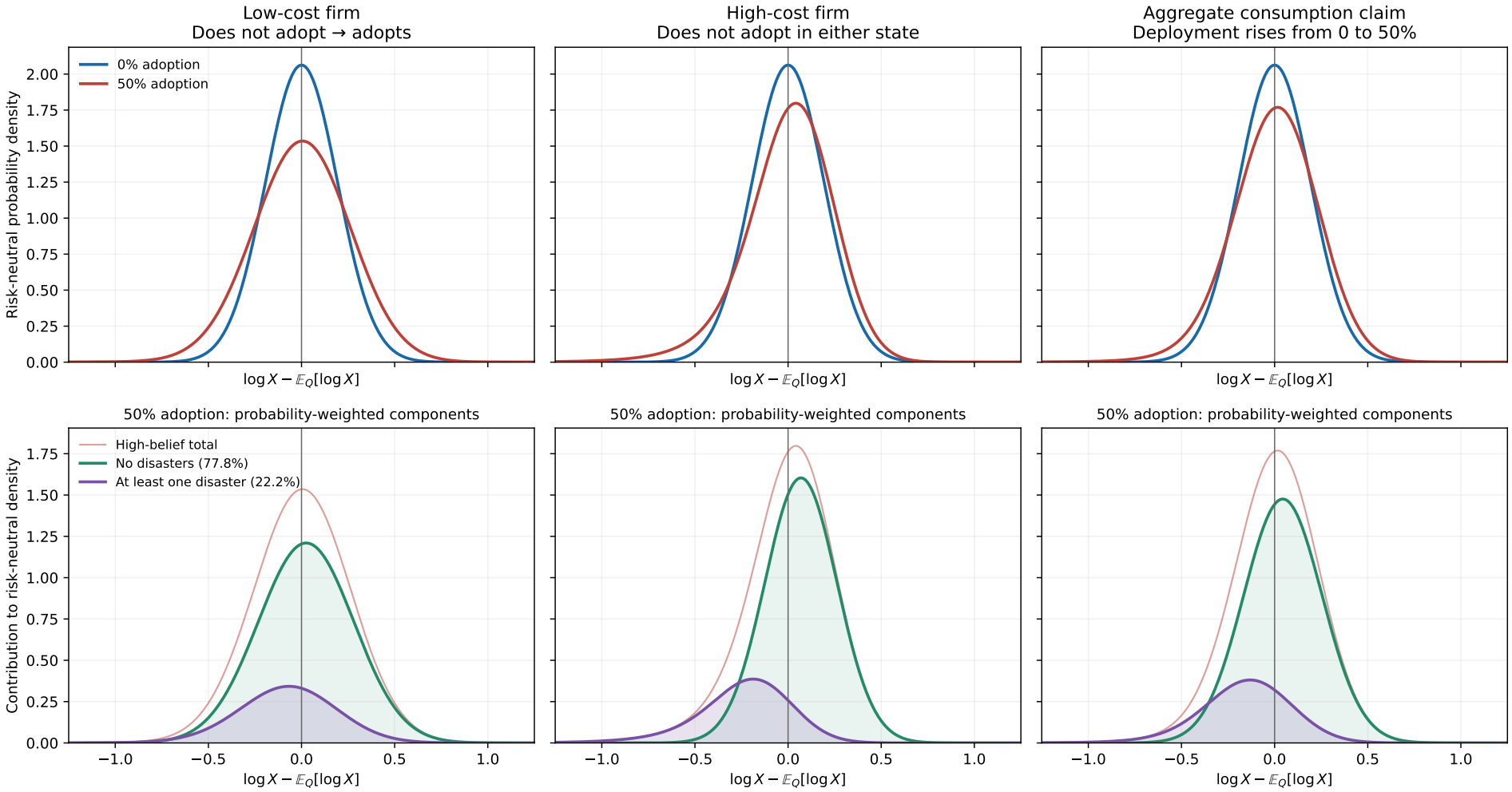


Figure 2: **Risk-neutral payoff densities as adoption rises from zero to** 50%**.** Columns show the low-cost firm, high-cost firm, and aggregate consumption claim. Each top-row density is centered at its own Q mean of log payoff. The bottom row decomposes the higher-belief density into probability-weighted no-disaster and disaster components. The components share the total density's center and integrate to their event probabilities. They sum to the red curve. The horizon is 16 years after review.

dates. This extension also makes the treatment of extinction consequential for private prices. We compare four cases: economy-wide survivable losses, adopter-only losses, extinction that ends the consumption problem, and extinction with a fixed reservation consumption level. All four use the same technology, adoption costs, and hazard schedule. We normalize year-four adoption to the same share in each case and compare its deployment and risk with a no-disaster economy receiving the same productivity news.

3.1 Dividends, adoption, and information

There is a unit mass of firms. Before adoption, each firm pays the old-technology dividend B_t, whose log growth, before technological disasters, is

$$d \log B_t = g \, dt + \sigma_O \, dW_t. \tag{21}$$

The new technology's relative productivity signal satisfies

$$dy_t = \psi \, dt + \sigma_N \, dZ_t. \tag{22}$$

Capability ψ is fixed but unknown and is measured in units of annual relative log-dividend growth; W_t and Z_t are independent Brownian motions. Investors observe the public signal y_t whether or not firms adopt, and update their beliefs about ψ as productivity news arrives. Good news raises expected capability, while uncertainty based on this signal alone declines deterministically with time. The public posterior distribution of capability, including disaster information described below, is π_t, with mean $\widehat{\psi}_t$. Adoption does not generate additional productivity observations or improve their precision. In the absence of disasters, neither an individual firm nor the planner can learn more by deploying to more firms; waiting allows learning from the ongoing public signal before committing to adoption.

Firm $u \in [0, 1]$ pays the same type-dependent conversion cost as before, $\kappa_\epsilon(u) = \kappa + \epsilon(2u - 1)$. Conversion permanently reduces its dividend scale by $1 - \kappa_\epsilon(u)$, after which it earns the new technology's relative growth. If firm u adopts at date τ_u, its dividend immediately after conversion is $D_{u,\tau_u} = [1 - \kappa_\epsilon(u)]B_{\tau_u}$. Between disaster arrivals, its dividend process is

$$d \log D_{u,t} = (g + \psi) \, dt + \sigma_O \, dW_t + \sigma_N \, dZ_t, \qquad t > \tau_u. \tag{23}$$

Thus an adopter receives the old-economy growth and shock plus the new technology's relative growth and shock. This dividend process simplifies the productivity dynamics of Pástor and Veronesi (2009). Instead of tracking a separate mean-reverting productivity state, the growth advantage ψ enters dividends immediately, and the technology shock dZ_t directly affects all adopters' dividend growth. The same observed process therefore supplies information about capability and determines adopters' relative dividend growth. Let q_t be the adopted mass and a_t the total dividend of adopted firms divided by B_t. Consumption is exactly the sum of dividends,

$$C_t = B_t(1 - q_t + a_t). \tag{24}$$

Writing $K(q) = \int_0^q [1 - \kappa_\epsilon(u)]du$, conversion from q to $r \geq q$ changes installed dividends to $a^+ = a + K(r) - K(q)$. The associated loss of consumption is $B \int_q^r \kappa_\epsilon(u)du$. Firms can wait before

adopting but cannot reverse adoption.

Disasters have intensity $q_t\lambda(\psi)$, where λ is bounded and increasing. Deployment exposure is linear, as in Section 2. Each survivable event draws a retained dividend fraction Ω. Economy-wide losses multiply all dividends by this fraction; adopter-only losses multiply only adopted firms' dividends. These disasters can recur. Extinction is absorbing and sets every financial payoff to zero.

Disaster observations provide a second source of learning. Since high capability raises the hazard, a survivable disaster raises inferred capability, while survival without a disaster lowers it relative to beliefs based on productivity alone. Aggregate deployment changes exposure to the hazard and hence the information conveyed by these observations. An individual firm takes that information process as given; the planner accounts for how deployment changes it. This is the only channel through which adoption affects learning. Appendix E gives the posterior and computational details.

3.2 Four disaster cases

Let $V_t > 0$ denote lifetime utility in consumption-equivalent units, γ risk aversion, and η the inverse elasticity of intertemporal substitution (IES). Set $\rho = 1-\eta$, let h be the decision interval, and write $\beta = e^{-\delta h}$ for discounting. The recursive preferences of Epstein and Zin (1989) take the form

$$V_t^\rho = hC_t^\rho + \beta Q_t^\rho, \qquad Q_t = \left(\mathbb{E}_t[V_{t+h}^{1-\gamma}]\right)^{1/(1-\gamma)}. \tag{25}$$

The certainty equivalent Q_t summarizes uncertainty about future utility. Risk aversion governs the valuation of this uncertainty; the IES governs substitution between current consumption and continuation. We use $\gamma = 4$ and IES 2, so $\eta = 1/2$ and $\rho = 1/2$. Consumption is paid through a finite horizon T, with no terminal lump-sum dividend.

Economy-wide survivable losses. Each disaster multiplies every firm's dividends by the same retained fraction Ω, drawn independently across events. Consumption falls proportionally, while the adopter–non-adopter payoff ratio is unchanged. Because disasters are more likely when capability is high, their marginal-utility cost raises the value of adopters' exposure to capability. Equation (25) includes both the consumption loss and the value of the surviving economy, and disasters can recur.

Adopter-only losses. Each disaster instead multiplies only adopted firms' dividends by the random retained fraction Ω. Consumption again falls, but adopters now lose relative to non-adopters precisely when the loss is most costly. This direct exposure works against the hedging incentive. Preferences and the hazard schedule are otherwise the same as in the economy-wide-loss case.

Extinction with end of consumption. Every financial claim pays zero after extinction. Let s_t be the probability of surviving the next interval and let

$$A_t = \left(\mathbb{E}_t[V_{t+h}^{1-\gamma} \mid \text{survival}]\right)^{1/(1-\gamma)} \tag{26}$$

be the certainty equivalent of surviving continuation value. This includes uncertainty about consumption and any subsequent extinction. Following the no-bequest mortality specification of Li and Smetters (2010), we weight continuation by survival probability outside the risk aggregator:

$$(V_t^{\text{end}})^\rho = hC_t^\rho + \beta s_t A_t^\rho. \tag{27}$$

Current consumption is received before the event; extinction ends the consumption problem. In Li and Smetters, survival is exogenous individual mortality. Here its probability depends on technological capability and deployment.

Extinction with fixed reservation consumption. All financial claims again pay zero, but extinction is assigned the utility of a fixed nonfinancial consumption flow $c_E = \underline{c}C_0$ through T. Let $D_{t+h} > 0$ denote the consumption-equivalent value of that remaining flow. Extinction is now an outcome inside the risk aggregator:

$$(V_t^{\text{res}})^\rho = hC_t^\rho + \beta(Q_t^{\text{res}})^\rho, \tag{28}$$

$$Q_t^{\text{res}} = \left[s_t A_t^{1-\gamma} + (1-s_t)D_{t+h}^{1-\gamma}\right]^{1/(1-\gamma)}. \tag{29}$$

The reservation flow is fixed at 60% of initial consumption. It does not depend on adoption, capability, or consumption when extinction occurs. Its value changes with the remaining horizon, but investors cannot purchase it through a financial claim.

We do not take a stand on which extinction specification is more appropriate. The adoption response in either case depends on parameters; the examples illustrate the balance between hedging and discounting rather than a fixed ranking of deployment across specifications. To see the distinction, ask what terminal value D_{t+h}^* inserted in (29) would reproduce (27). Equating their continuation terms gives

$$\left[s_t A_t^{1-\gamma} + (1-s_t)(D_{t+h}^*)^{1-\gamma}\right]^{\rho/(1-\gamma)} = s_t A_t^\rho. \tag{30}$$

For a short interval with a small extinction probability,

$$\frac{D_{t+h}^*}{A_t} \longrightarrow \left(\frac{\gamma-\eta}{1-\eta}\right)^{1/(1-\gamma)} = 7^{-1/3} \simeq 0.523. \tag{31}$$

Thus end of consumption implicitly assigns extinction a value proportional to surviving continuation value, approximately 52.3% at our parameters. This is a utility-equivalent representation, not

consumption or wealth received after extinction. Appendix F derives the exact mapping and its pricing implications.

The distinction can also be expressed as willingness to pay for survival. Martin and Pindyck (2021) express catastrophic loss of life as a welfare-equivalent reduction in consumption, calibrated using estimates of the value of a statistical life. Holding disaster probabilities and consumption growth fixed, end of consumption implies that investors would surrender the same fraction of their entire consumption stream to eliminate extinction risk as society becomes richer. With a reservation stream fixed in consumption units, that fraction rises with prosperity: the fall to the same living standard becomes relatively more costly. This connects the comparison to the value-of-life literature on willingness to pay for safety as income rises (Hall and Jones, 2007; Jones, 2016). Under CRRA, changing the extinction valuation affects welfare without changing private prices at a given allocation. Recursive preferences make the value assigned to continued existence consequential for financial choices (Bommier et al., 2023); here it changes the marginal valuation of surviving payoffs.

3.3 Hedging, discounting, and adoption

Let S_{t+h} indicate survival. A claim paying X_{t+h} on survival and zero on extinction has price $\mathbb{E}_t[S_{t+h}M_{t,t+h}X_{t+h}]$. Its surviving-state pricing kernel is

$$M_{t,t+h} = \beta \left(\frac{C_{t+h}}{C_t}\right)^{-\eta} \left(\frac{V_{t+h}}{Q_t^j}\right)^{\eta-\gamma}, \qquad Q_t^j = \begin{cases} A_t, & \text{end of consumption,} \\ Q_t^{\text{res}}, & \text{fixed reservation consumption.} \end{cases} \tag{32}$$

For survivable disasters the same formula uses Q_t from (25), without a survival indicator. Consumption and continuation value incorporate aggregate adoption. Under CRRA, $\eta = \gamma$, so the continuation-value term disappears. This is why the fixed-date model can separate private pricing from the utility assigned to extinction.

With recursive preferences, news about future disaster risk also affects prices through continuation value. Since $\eta - \gamma < 0$, a reduction in future utility raises the pricing weight on the state in which that news arrives. Adopters' exposure to capability can therefore hedge news about extinction risk even though no firm pays anything after extinction. A low fixed reservation value can strengthen this hedging incentive by making the threatened loss of future utility more severe.

Adoption, however, exchanges current consumption for future dividends. What matters is both the value of adopters relative to other firms and the value of future financial payments relative to consumption today. The price and yield of a bond paying one only on survival are

$$B_t^S = \mathbb{E}_t[S_{t+h}M_{t,t+h}], \qquad y_t^S = -\frac{\log B_t^S}{h}. \tag{33}$$

These are survival-contingent yields in the extinction cases; in the survivable cases a risk-free bond pays in every state.

Higher capability affects discounting through three channels. First, higher expected consumption growth raises the incentive to move consumption toward the present. The IES governs this growth effect separately from risk aversion. Second, a higher extinction probability reduces the chance that a financial payment will arrive. Both extinction specifications retain this physical-survival effect, which underlies the interest-rate mechanism in Chow et al. (2026). Third, under fixed reservation consumption, a more valuable surviving future widens the proportional gap between survival and extinction. Investors become less willing to sacrifice consumption today for payments received only in the comparatively good survival state. This depresses the prices of bonds and investments in the new technology alike.

The last two channels can be separated algebraically. Let λ_t denote local extinction intensity, so $s_t = e^{-\lambda_t h}$. Holding surviving consumption and continuation prospects fixed, the direct effect on the short yield is, to first order in h,

$$\frac{\partial y_t^{\text{end}}}{\partial \lambda_t} = 1, \tag{34}$$

$$\frac{\partial y_t^{\text{res}}}{\partial \lambda_t} = 1 + \frac{\gamma - \eta}{\gamma - 1}\left[\left(\frac{A_t}{D_{t+h}}\right)^{\gamma-1} - 1\right]. \tag{35}$$

End of consumption retains the physical-probability adjustment. Holding surviving prospects fixed, fixed reservation consumption adds a positive discounting effect when $D_{t+h} < A_t$ and $\gamma > \eta$, as in our calibration. This algebraic comparison does not depend on the simulated adoption paths: the additional effect grows with the gap between surviving and extinction values. With $\gamma = 4$, IES 2, and $D_{t+h}/A_t = 1/2$, the direct yield sensitivity is approximately 9.2, compared with one under end of consumption. These are local pricing effects; the simulations incorporate the associated changes in adoption and future risk.

The reservation specification therefore need not weaken the hedge to reduce adoption. It can strengthen the hedge while strengthening discounting even more. End of consumption can instead retain a strong hedging incentive with a smaller discounting response. The resulting adoption choices feed back into disaster exposure. The planner uses the same preferences and consumption accounting, but internalizes deployment's effects on both disaster risk and future information.

3.4 Illustrations at a common adoption normalization

The quantitative exercise illustrates how much disaster risk can change adoption. All four cases use the same technology, uncertainty, conversion costs, and disaster hazard schedule; they differ in who bears the losses and how extinction enters utility. Adoption costs average 10%, and the technology offers substantial gains: expected dividend growth for adopters is 1.18% to 2.48% higher annually than the old technology's 2.5% growth rate at the year-four comparison point. Survivable disasters now have a range of possible sizes, rather than the fixed loss in Section 2. Losses average 22.3%, with occasional larger declines. Table 2 reports the parameters, and Appendix E gives the distribution and numerical details.

Table 2: Dynamic parameters and capability beliefs at the adoption normalization

Risk aversion γ	4	IES $1/\eta$	2
Discount rate δ	0.03	Old log growth g	0.025
Old volatility σ_O	0.02	Signal volatility σ_N	0.03
Prior standard deviation	0.014204	Hazard location $\psi_\star$	0.010561
Mean conversion cost κ	0.10	Cost half-width ϵ	0.015
Hazard scale	2.25	Extinction reservation flow	$0.60C_0$
Mean disaster loss	22.3%	Horizon T	30 years

Case matching 56.1% adoption at year four	$\widehat{\psi}_4$
Economy-wide survivable loss	0.01584
Adopter-only loss	0.02484
Extinction: end of consumption	0.01181
Extinction: fixed reservation	0.02170

The subjective mean $\widehat{\psi}_4 = \mathbb{E}[\psi \mid \mathcal{F}_4]$ incorporates both productivity observations and survival without a disaster. It is the belief about capability; expected relative dividend growth is $\widehat{\psi}_4$.

For each disaster case, we choose a path of favorable productivity news that brings adoption to 56.1% after four years. This is the latest Ramp share reported in Section 4, used as a common scale for the exercise rather than a precise match between survey use and model adoption. We then ask how much firms would adopt after the same news if there were no disaster risk, and how much a planner would choose. These comparisons show whether market pricing accelerates or restrains adoption and how the resulting deployment compares with the social optimum.

No disaster occurs along the displayed paths. Because each case requires different news to reach the adoption target, the relevant comparison is within each case: the market, no-disaster benchmark, and planner all receive the same productivity news.

Figure 3 shows the two survivable cases. With economy-wide losses, year-four adoption is 56.1%, compared with 35.0% without disaster risk. Perceived annual risk is 1.27%, compared with 0.80% if deployment followed the no-disaster path. The hedging incentive therefore raises both deployment and perceived danger. In the adopter-only case, the productivity news needed to reach the same adoption target would produce 100.0% deployment without disaster risk. Direct exposure to the loss reduces adoption to 56.1% and perceived risk from 4.06% under no-disaster deployment to 2.34%. The planner chooses 0.0% adoption on the economy-wide-loss path and 43.4% on the adopter-only path.

Figure 4 shows how the treatment of extinction changes this comparison. With end of consumption, productivity news sufficient to produce 56.1% adoption generates 3.3% adoption by year four in the no-disaster economy. Perceived annual extinction risk is 0.94%, and the survival-contingent yield is approximately 4.69%, compared with 3.71% without disaster risk. The hedging incentive dominates discounting and brings adoption forward.

With fixed reservation consumption, stronger productivity news than under end of consumption is needed to attain the same adoption share. Those observations would produce 86.5% adoption

without disaster risk. Perceived annual extinction risk is 1.91%, compared with 2.90% under no-disaster deployment. The survival-contingent yield reaches approximately 14.51%, against 3.19% without disaster risk. The large yield response is the quantitative expression of the additional discounting effect in (35). At year four, this discounting effect dominates: adoption is 30.3 percentage points below the no-disaster benchmark. At year eight, adoption is 60.0% with extinction risk and 100.0% without it. The planner chooses no deployment on either displayed extinction path.

The four cases separate two ways markets can discipline adoption. Adopter-only losses reduce the technology's relative payoff in bad states. Fixed reservation consumption instead makes surviving future payments collectively less attractive relative to current consumption. End of consumption preserves the physical-survival discounting effect but avoids this additional fixed-reservation penalty. In this illustration, economy-wide survivable risk and end-of-consumption extinction bring adoption forward, while adopter-only losses restrain it. Fixed-reservation extinction briefly brings initial adoption forward, but its stronger discounting effect reduces deployment by year four and thereafter.

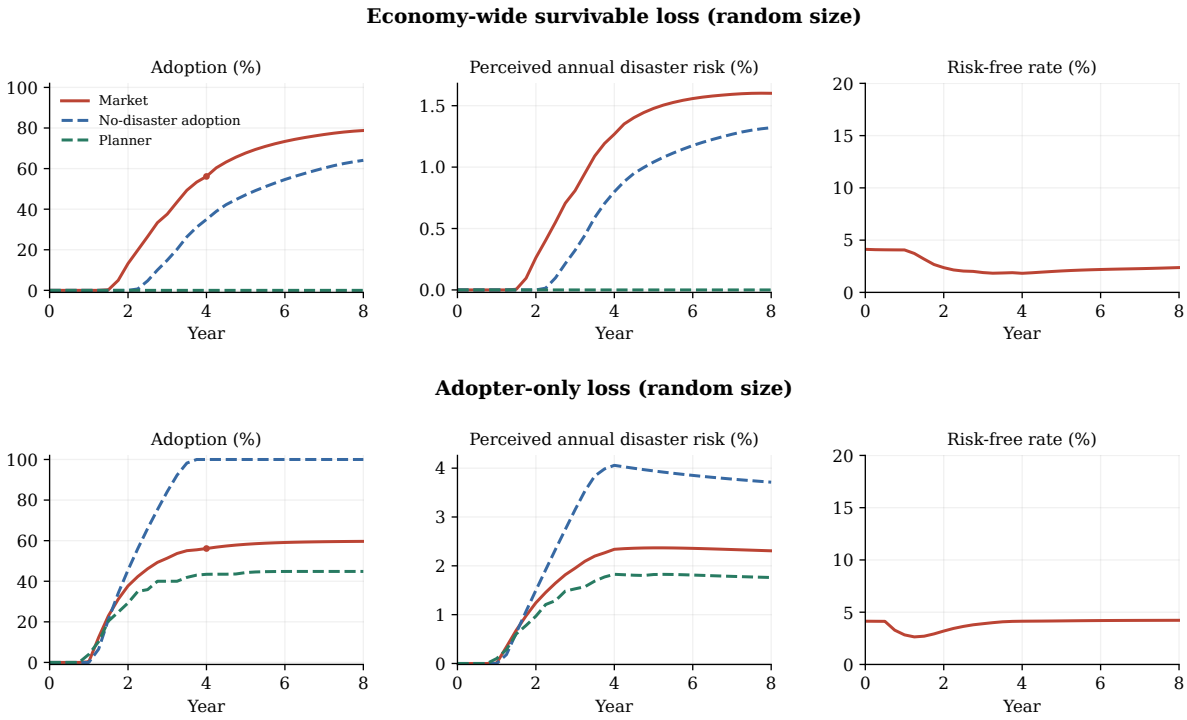


Figure 3: **Survivable disasters at a common adoption normalization.** The upper row shows economy-wide losses and the lower row adopter-only losses. Both use the same random loss distribution, with mean 22.3%. In each row, productivity news is chosen to match market adoption of 56.1% at year four and is identical for the market, planner, and no-disaster comparison. Solid red curves show the disaster-economy market allocation; dashed green and blue curves show planner and no-disaster adoption benchmarks. Blue risk applies the disaster hazard and survival updating to no-disaster adoption. Risk is the perceived probability of at least one disaster over a year at current deployment. The rate panels show only the market path. For readability, the yield curves smooth the component associated with quarterly conversion costs; Appendix E describes the procedure.

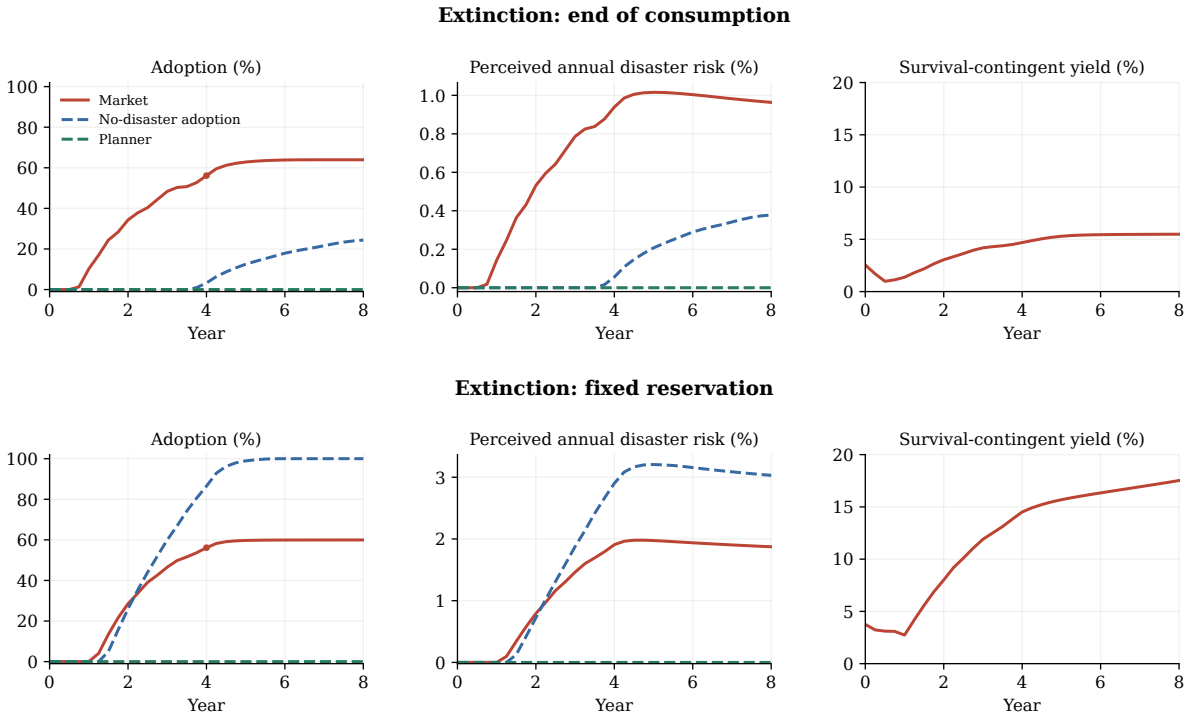


Figure 4: **Extinction at a common adoption normalization.** The upper row ends the consumption problem; the lower row assigns a nonfinancial reservation flow of 60% of initial consumption. Every financial claim pays zero after extinction. Productivity news is matched separately in each row to market adoption of 56.1% at year four, and the market, planner, and no-disaster comparisons within that row receive identical news. Colors and perceived risk follow Figure 3. For readability, the yield curves smooth the component associated with quarterly conversion costs; Appendix E describes the procedure. Extinction-case yields are on bonds that pay only upon survival.

3.5 Technology news and option-implied tails

At year four, we match deployment to 56.1% separately in each economy, as in the time-series figures, and introduce a positive, one-standard-deviation monthly productivity innovation. Figure 5 compares the market, an adopted firm, and the highest-cost firm, which has not yet adopted. This firm retains the option to adopt later at cost $\kappa + \epsilon$. Survivable losses follow the same random-size distribution as in the time-series illustrations.

We measure the cumulative probabilities that the stock price one year ahead is at least 10% below or above each claim's forward.[4] Updating the forward after the news isolates changes in distributional shape from proportional price changes. The model probabilities include all outcomes beyond each threshold; Appendix E.4 gives the calculation.

The economy-wide survivable case provides the prediction we take to the data. Positive news about the new technology raises both downside and upside tail probabilities for the market and adopted firms. The old-technology firm's response is more asymmetric: its downside probability rises by 1.30 percentage points, compared with 0.68 points for its upside probability. Thus favorable technology news can increase risk in both tails while producing a stronger downside response among firms that have not adopted.

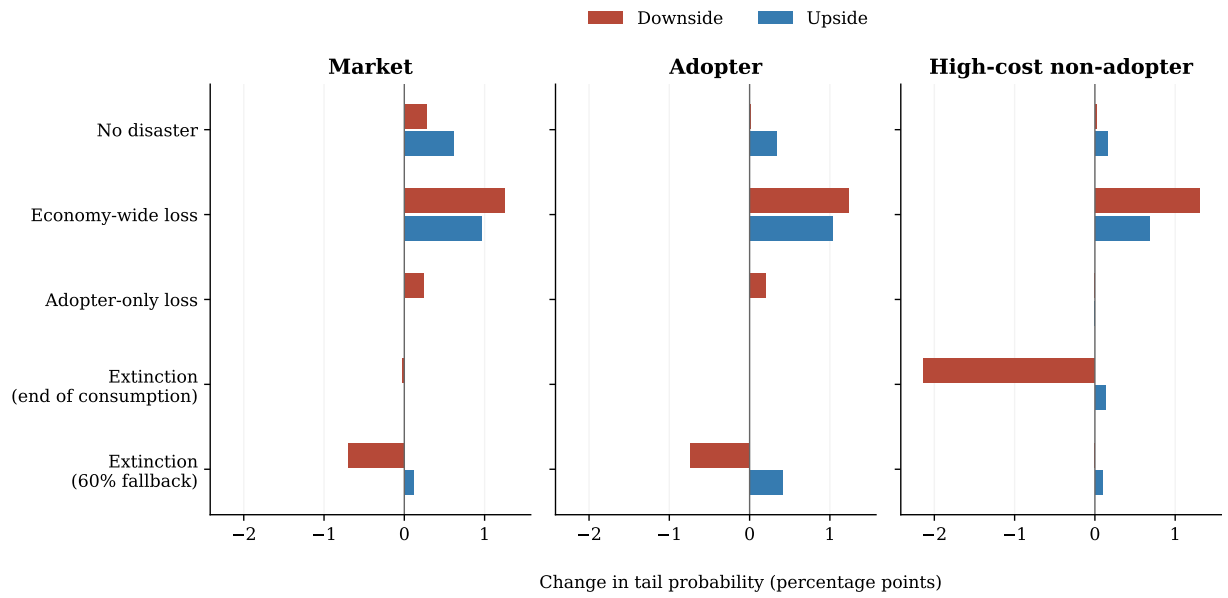


Figure 5: **Technology news and option-implied tails at** 56.1% **deployment.** Bars show percentage-point changes in one-year downside and upside probabilities following positive productivity news at year four. The cumulative probabilities include all stock prices at least 10% below or above the claim's own forward, updated after the news. Survivable disasters have random loss sizes; reservation consumption is 60% of initial consumption. The non-adopter is the highest-cost firm and retains its future adoption option. Extinction-case option probabilities condition on financial survival.

[4]Section 4 uses strikes 20% from the forward. The magnitudes and relative tail responses depend on moneyness.

The other disaster specifications produce different responses. Adopter-only losses leave the old-technology firm's tails nearly unchanged. Under end-of-consumption extinction, adoption responds strongly to productivity news, increasing the likelihood that even the highest-cost firm adopts within the following year. Its adoption option cushions the left tail by allowing it to benefit from higher productivity in states where rising discount rates would otherwise depress the value of its old-technology dividends. Its downside probability consequently falls. Fixed reservation consumption reduces downside probabilities for the market and adopted firms, while leaving the non-adopter's downside probability nearly unchanged.

The index-option tests in Section 4 examine the economy-wide survivable prediction: whether positive technology-stock returns accompany increases in both tail probabilities, with a stronger downside asymmetry for the old-economy portfolio. Several forces can generate these patterns, so we view the evidence as suggestive.

4 Technology Stocks and Tail Risks in the Data

The dynamic model motivates our empirical tests. In the economy-wide survivable case, positive technology news raises both one-year option-implied tails, with a stronger downside asymmetry for the non-adopter (Figure 5). We examine one-year option-implied tail probabilities for three equity portfolios represented by exchange-traded funds (ETFs): Invesco QQQ (QQQ), which tracks the technology-heavy Nasdaq-100 Index; the iShares Russell 2000 ETF (IWM), which tracks small-cap U.S. stocks; and the SPDR S&P 500 ETF Trust (SPY), which tracks the large-cap U.S. market. We select these ETFs for their highly liquid options markets: SPY, QQQ, and IWM rank first, second, and third in ETF options trading volume.[5] QQQ represents technology-exposed, relatively low-cost adopters, while IWM represents relatively high-cost adopters. QQQ returns proxy for changes in the value of technology-intensive firms.

The mapping draws on the greater AI use among large firms documented by Bonney et al. (2026) and differences in firms' labor-task exposure. Using the exposure measure of Eisfeldt et al. (2023), we compare firms in the two extreme groups: artificial (A), whose workforce tasks are most exposed to AI, and human (H), whose tasks are least exposed. These are the top and bottom fifths of the exposure ranking. Among firms in these two groups, 41.1% of IWM firms are H and 58.9% are A, compared with 25.7% H and 74.3% A for QQQ. IWM thus has a more human-oriented task composition by firm count. Labor-task exposure thus provides a useful, though imperfect, proxy for the distinction between adopters and non-adopters in the model. Appendix H gives the classification details.

[5]The ranking uses average daily contract volume across the Cboe options exchanges in 2026 through August 28, as compiled by Macroption. For the S&P 500 tail probabilities, we use SPX index options and pair them with SPY returns.

4.1 Tail probabilities and the rolling specification

Let j index the equity series and t index month-end dates. Write $S_{j,t}$ for the underlying index or ETF price, $F_{j,t}$ for its one-year forward price, and $\tau = 365/365$ years for maturity. Let Q_t denote the risk-neutral probability measure conditional on date-t information. The downside and upside objects are approximately

$$p^-_{j,t} = Q_t(S_{j,t+\tau} < 0.8F_{j,t}), \qquad p^+_{j,t} = Q_t(S_{j,t+\tau} > 1.2F_{j,t}). \tag{36}$$

We estimate the probabilities from option spreads centered on these thresholds. Write $F = F_{j,t}$ and suppress the index and maturity arguments. If $\mathcal{P}(k)$ and $\mathcal{C}(k)$ denote put and call prices at strike k, and d_t is the discount factor to option expiry, the maturity-specific estimators are

$$\widetilde{p}^-_{j,t} = \frac{\mathcal{P}(0.85F) - \mathcal{P}(0.75F)}{d_t\, 0.10F}, \tag{37}$$

$$\widetilde{p}^+_{j,t} = \frac{\mathcal{C}(1.15F) - \mathcal{C}(1.25F)}{d_t\, 0.10F}. \tag{38}$$

These are strike-band averages of digital probabilities. Available expirations are interpolated to one year; month-end observations are selected separately by wing and daily returns compounded monthly. Changes and returns are in percentage points, so a coefficient measures the probability change per percentage-point QQQ return, controlling for SPY.

For index j and tail $s \in \{-, +\}$ (downside or upside), let w be a window of month-end observations. Let r_t^{QQQ} and r_t^{SPY} denote monthly QQQ and SPY returns. In the regression, $\Delta\widetilde{p}^s_{j,t}$ is the monthly change in the estimated probability, $\alpha^s_{j,w}$ is the intercept, $\beta^s_{j,w}$ and $\theta^s_{j,w}$ are the two return slopes, and $\varepsilon^s_{j,t}$ is the residual. Each window estimates

$$\Delta\widetilde{p}^s_{j,t} = \alpha^s_{j,w} + \beta^s_{j,w} r_t^{QQQ} + \theta^s_{j,w} r_t^{SPY} + \varepsilon^s_{j,t}, \qquad t \in w. \tag{39}$$

Each window uses 13 consecutive months with the same intercept and QQQ/SPY return controls. Figure 6 plots $\beta^s_{j,w}$ and pointwise normal 95% Newey–West intervals with three lags. QQQ's coefficient includes an own-return relation. SPY-labelled probabilities come from SPX options.

The history combines OptionMetrics and a Databento extension with the established vendor adjustments. Appendix H describes the splice and its measurement limitations. The displayed windows end from February 2023 through August 2026. The dashed Ramp series reports AI use by businesses, reaching 56.1% in August 2026, the share used to normalize adoption in Section 3. It uses a separate right-hand axis and is not included in the regressions.

4.2 Findings and interpretation

Figure 5 provides the model counterpart to these empirical tail responses. In the economy-wide survivable case, favorable capability news increases both downside and upside probabilities measured relative to each claim's updated one-year forward. The non-adopter's downside response exceeds

its upside response. We therefore examine whether favorable technology-stock returns coincide with increases in both option-implied tails and a stronger downside asymmetry for the old-economy portfolio. Technology returns serve as a proxy for favorable capability news.

Figure 6 shows that favorable QQQ returns are associated with increases in both broad-market and QQQ tail probabilities in the later windows. QQQ's upside response is more precisely estimated than its downside response. For IWM, the pattern is more asymmetric: the downside response is significant in the later period, while the upside response is mostly insignificant. The shift from negative downside coefficients in early windows to positive coefficients later contrasts with the usual leverage intuition, under which falling stock prices raise risk. Here, favorable technology returns increasingly coincide with greater downside risk.

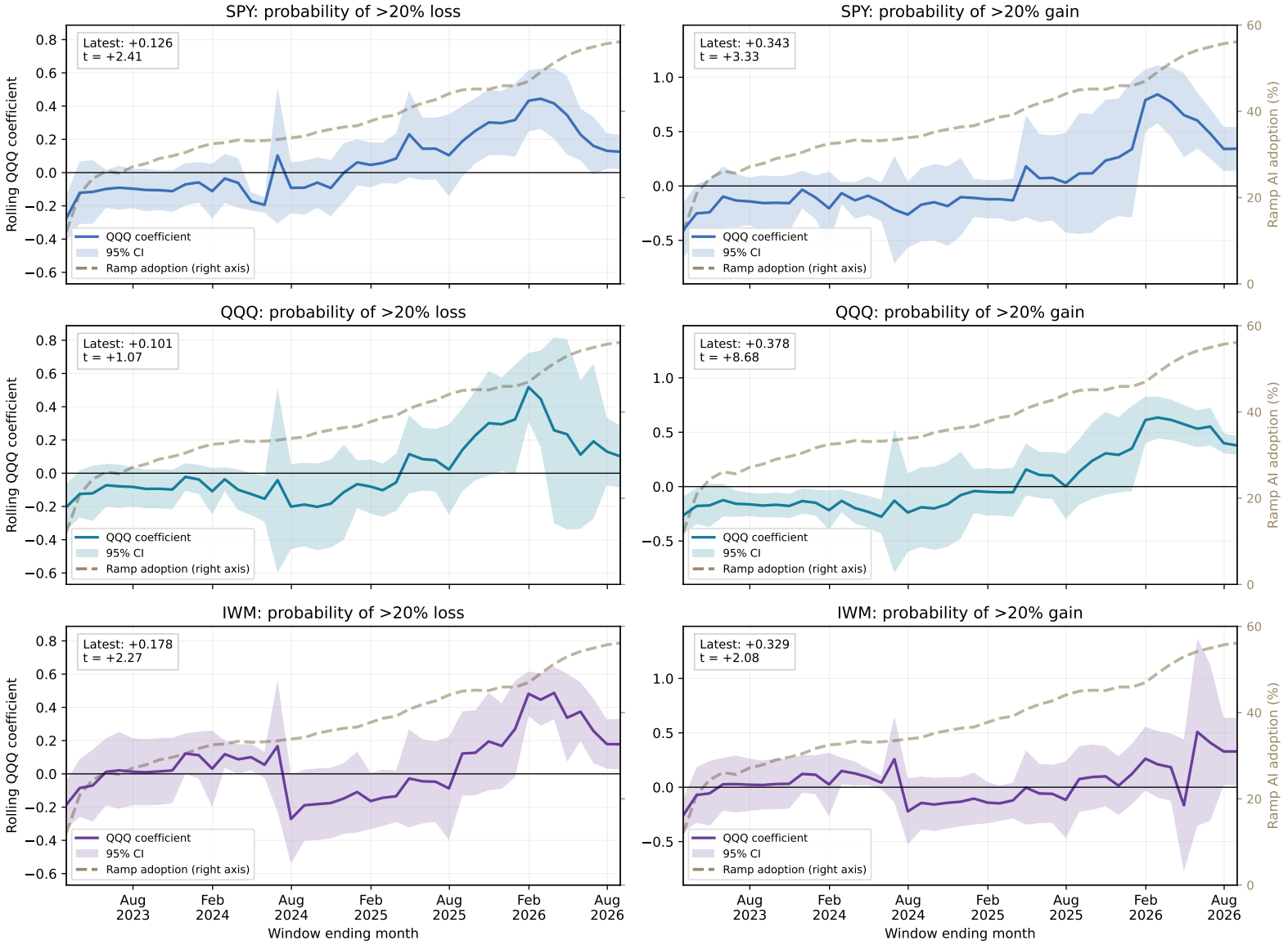


Figure 6: **Rolling QQQ-return coefficients for one-year downside and upside probabilities.** Rows show the broad market (SPX probabilities, labeled SPY), QQQ, and IWM. Columns show loss and gain probabilities beyond 20% of the one-year forward. Each 13-month window estimates $\Delta\widetilde{p}^{s}_{j,t} = \alpha^{s}_{j,w} + \beta^{s}_{j,w} r^{QQQ}_{t} + \theta^{s}_{j,w} r^{SPY}_{t} + \varepsilon^{s}_{j,t}$, where $\Delta\widetilde{p}^{s}_{j,t}$ is the monthly change in the tail probability and the returns are monthly QQQ and SPY returns. Curves show $\beta^{s}_{j,w}$ against the window-ending month. Shading gives pointwise 95% Newey–West intervals with three lags. With only 13 monthly observations per fit, these asymptotic intervals and the reported t-statistics are approximate. The dashed Ramp adoption level uses the right axis and is not included in the regressions. February 2023–August 2026.

The contrast between QQQ and IWM is consistent with firms differing in their ability to capture technology gains while sharing exposure to its risks. Technology returns and option-implied tails also reflect broader growth expectations and risk appetite, and the recent emergence of generative AI leaves a short history for estimation. We therefore view the findings as suggestive evidence for the model's central empirical prediction: favorable technology news can coincide with greater downside risk, including for firms less exposed to the technology's productivity benefits.

5 Conclusion

Markets can reward the adoption of a technology that makes a disaster more likely even when adopting and non-adopting firms suffer the same proportional loss. When capability raises both productivity and danger, disasters arrive disproportionately in the states where adopters are relatively most productive, and those are states of high marginal utility. Private firms price that payoff pattern while taking the aggregate hazard as given, so disaster risk raises the private return to adoption above what it would be if the technology were safe. A planner, internalizing the hazard that deployment creates, moves the other way. Market pricing can therefore amplify the deployment externality by increasing the private return to adoption.

The nature of the calamity determines whether this happens. A economy-wide survivable loss cancels from the adopter–non-adopter payoff ratio and survives only in state prices, where its dependence on capability encourages adoption. Damage confined to adopters is a direct relative cash-flow loss that competes with the selection benefit and restrains adoption in our benchmark. In the simple CRRA model, extinction removes high-capability payouts and modestly restrains adoption through physical survival probabilities. With recursive preferences, hazard news also changes the value of surviving claims. Ending the consumption problem can encourage adoption, while a low fixed reservation utility raises survival-contingent yields and can substantially restrain it.

The mechanism relies on danger rising with capability. If advances in control and monitoring arrive alongside advances in capability, the selection effect weakens, and if danger falls with capability it reverses. Measures that reduce the hazard itself therefore weaken the hedging incentive along with the danger, while the incentive is strongest exactly when the technology is most dangerous.

These results suggest several policy responses. A corrective deployment charge can make firms internalize the additional hazard they create. An industry-wide penalty conditional on a survivable disaster could instead reduce the relative payoff to adoption in disaster states, provided liability is linked to each firm's deployment and the proceeds are transferred rather than destroyed. Such penalties cannot be collected after extinction. Measures that reduce disaster likelihood or severity offer another route, lowering expected damage and, under the conditions of Proposition 1, weakening the private hedging incentive.

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A Fixed-date threshold proofs

A.1 Private threshold

Proof. Recall that H is aggregate terminal wealth without disasters, $H_q = \partial H/\partial q$ is the marginal conversion payoff, γ is risk aversion, and R_D is the economy-wide-disaster valuation multiplier. To see the sign, let P denote the physical probability measure and Q_0 the no-disaster state-price measure. Evaluated at the no-disaster threshold, its density is

$$\frac{dQ_0}{dP} = \frac{H^{-\gamma}}{\mathbb{E}[H^{-\gamma}]}. \tag{40}$$

At that threshold $\mathbb{E}^{Q_0}[H_q] = 0$. Both $R_D(q,\psi)$ and $H_q(q,\psi)$ are increasing in ψ, so for $q > 0$,

$$\mathbb{E}^{Q_0}[R_D H_q] = \mathrm{Cov}^{Q_0}(R_D, H_q) > 0. \tag{41}$$

The marginal firm therefore prefers adoption at the old threshold, and single crossing implies a lower threshold. At $q = 0$, $\Lambda(0,\psi) = 0$ and $R_D(0,\psi) = 1$.

For the comparative statics, compare two economy-wide-disaster economies with multipliers R_1 and R_2, holding q, the posterior variance, and all payoff primitives fixed. A lower retained fraction raises $\chi = \omega^{1-\gamma} - 1$. A proportional increase in intensity has the same effect on the coefficient of $\lambda(\psi)$ in the exponent. In either case,

$$\frac{R_2(q,\psi)}{R_1(q,\psi)} = \exp\{c\lambda(\psi)\}, \qquad c > 0. \tag{42}$$

At economy 1's threshold, define its normalized valuation measure by

$$\frac{dQ_1}{dP} = \frac{R_1 H^{-\gamma}}{\mathbb{E}[R_1 H^{-\gamma}]}. \tag{43}$$

Then $\mathbb{E}^{Q_1}[H_q] = 0$, while

$$\mathbb{E}^{Q_1}\left[\frac{R_2}{R_1} H_q\right] = \mathrm{Cov}^{Q_1}\left(\frac{R_2}{R_1}, H_q\right) > 0. \tag{44}$$

Strictness follows from nondegenerate capability uncertainty and strict monotonicity of both factors. Economy 2 therefore has a positive adoption gain at economy 1's threshold; its single-crossing condition implies a strictly lower threshold. At $q = 0$ both multipliers equal one, so the comparative effect vanishes. An additive capability-independent intensity increase instead makes R_2/R_1 constant in capability and leaves the threshold unchanged. All comparisons require finite valuation expectations.

□

A.2 Planner wedge

Proof. Recall that G_D^C and G_P^C are the decentralized and planner marginal gains, $\chi = \omega^{1-\gamma} - 1$ is the disaster valuation increment, and Δ is the remaining horizon. Subtracting the private condition from the planner condition gives

$$G_P^C - G_D^C = -\mathbb{E}\left[R_D H^{1-\gamma} \frac{\lambda(\psi)\Delta\chi}{\gamma - 1}\right] < 0. \tag{45}$$

Every factor inside the expectation is positive. The wedge is the marginal increase in the economy-wide disaster hazard created by additional deployment, which the firm takes as given and the planner internalizes. □

B Learning and pricing details

B.1 Productivity processes

The fixed-date economy has horizon $[0, T]$. A representative household owns firms, reinvests output before T, and consumes terminal wealth. Old-technology capital B_t^O grows at productivity ρ_t^O. Its long-run mean is $\bar{\rho}$, its mean-reversion speed is ϕ, and its innovation volatility is σ_O. Let $Z_{0,t}$ be the common standard Brownian motion:

$$dB_t^O = \rho_t^O B_t^O \, dt, \tag{46}$$

$$d\rho_t^O = \phi(\bar{\rho} - \rho_t^O) \, dt + \sigma_O \, dZ_{0,t}. \tag{47}$$

Productivity returns toward its long-run mean $\bar{\rho}$ at speed ϕ. The common old-economy shock is $Z_{0,t}$.

At t_0, investors begin observing a pilot sector using the new technology. Its long-run productivity is $\bar{\rho} + \psi$, with prior quality $\psi \sim \mathcal{N}(\mu_J, \sigma_J^2)$, where μ_J and σ_J are the prior mean and standard deviation. The pilot is small enough to reveal information without contributing materially to aggregate wealth. Write ρ_t^N for its productivity and $Z_{1,t}$ for an independent standard Brownian motion. Its loadings on the common and pilot-specific shocks are $\sigma_{N,0}$ and $\sigma_{N,1}$, respectively:

$$d\rho_t^N = \phi(\bar{\rho} + \psi - \rho_t^N) \, dt + \sigma_{N,0} \, dZ_{0,t} + \sigma_{N,1} \, dZ_{1,t}. \tag{48}$$

The second shock, $Z_{1,t}$, is independent of the old-economy shock. After subtracting observed common shocks and known mean reversion, investors obtain a noisy signal about ψ. The resulting posterior is Gaussian. Its mean $\widehat{\psi}_t$ responds to news, and its variance $\widehat{s}_t^2$ falls as observations accumulate. The signal and filter are given below.

Firms review adoption at t^*. We write $\widehat{\psi} = \widehat{\psi}_{t^*}$ and $\widehat{s}^2 = \widehat{s}^2_{t^*}$, so

$$\psi \mid \mathcal{F}_{t^*} \sim \mathcal{N}(\widehat{\psi}, \widehat{s}^2). \tag{49}$$

The fixed-date comparison isolates the uncertain long-run productivity gain. Write ρ_t^A for an adopter's productivity. It retains the old-economy shock but raises its long-run mean to $\bar{\rho} + \psi$:

$$d\rho_t^A = \phi(\bar{\rho} + \psi - \rho_t^A) \, dt + \sigma_O \, dZ_{0,t}, \qquad t \in [t^*, T]. \tag{50}$$

In the dynamic model, adopters receive dividend growth proportional to capability and are exposed to the shared frontier shock.

B.2 The fixed-date productivity signal

The coefficient $\sigma_{N,0}$ is exposure to the observable old-economy shock $Z_{0,t}$; $\sigma_{N,1} > 0$ is exposure to the new-technology-specific Brownian motion $Z_{1,t}$, which is independent of $Z_{0,t}$. Thus observed old- and new-sector productivity reveal a noisy signal about ψ. After removing the common shock

and known mean-reversion terms, the signal can be written

$$dY_t = \psi\, dt + \frac{\sigma_{N,1}}{\phi}\, dZ_{1,t}, \tag{51}$$

where Y_t is the cumulative orthogonalized productivity signal.

Let $\mathcal{F}_t$ denote the public information generated by observed old- and new-sector productivity through date t. Gaussian filtering implies

$$\psi \mid \mathcal{F}_t \sim \mathcal{N}(\widehat{\psi}_t, \widehat{s}_t^2), \qquad \widehat{s}_t^2 = \left[\frac{1}{\sigma_J^2} + \frac{\phi^2}{\sigma_{N,1}^2}(t - t_0)\right]^{-1}. \tag{52}$$

Here $\widehat{\psi}_t$ is the posterior mean and $\widehat{s}_t^2$ is the posterior variance. The posterior mean is a martingale that responds to unexpected new-technology productivity, whereas the posterior variance falls deterministically as signals accumulate. Equivalently, if $\widetilde{Z}_{1,t}$ denotes the innovation Brownian motion,

$$d\widehat{\psi}_t = \widehat{s}_t^2 \frac{\phi}{\sigma_{N,1}}\, d\widetilde{Z}_{1,t}. \tag{53}$$

Favorable AI news is therefore a positive innovation $d\widetilde{Z}_{1,t}$ and an upward revision in expected technology quality.

C Fixed-date incidence and welfare derivatives

C.1 Adopter-only damage

Recall that C_n^A is aggregate consumption after n adopter-only disasters and $U(C) = C^{1-\gamma}/(1-\gamma)$. Write $\Lambda_q = \partial\Lambda/\partial q$ for the marginal integrated hazard. In addition to the private gain G_D^A, the planner marginal gain G_P^A includes damage imposed on existing adopters:

$$G_P^A = G_D^A + \mathbb{E}_\psi\big[\Lambda_q \mathbb{E}_N\{U(C_{N+1}^A) - U(C_N^A)\}\big] = 0, \qquad \Lambda_q = \lambda(\psi)\Delta. \tag{54}$$

Here $\mathbb{E}_N$ conditions on quality ψ, and $\mathbb{E}_\psi$ averages over its posterior. Because $C_{N+1}^A < C_N^A$, the additional planner term is negative.

C.2 Extinction payoffs and the planner

Recall that S is the survival indicator and $H(q,\psi)$ is surviving terminal wealth. Utility is evaluated at an exogenous reservation-consumption floor $\underline{C} > 0$ after extinction,

$$C_T^E = SH(q,\psi) + (1-S)\underline{C}. \tag{55}$$

For $q > 0$, define $\bar{k}(q) = K(q)/q$ as the average cost-adjusted installed scale per adopter. Let $X_{O,T}^E$ and $X_{N,T}^E$ denote the terminal payoffs of an old-technology firm and the average new-technology adopter, respectively, in the extinction economy. Corporate payoffs are zero after extinction:

$$X_{O,T}^E = S, \qquad X_{N,T}^E = S\bar{k}(q)e^{A_2(\Delta)\psi}. \tag{56}$$

The reservation endowment is not corporate property. It is only a reduced-form convention that keeps welfare finite in the extinction state.

Let $s(q,\psi) = e^{-\Lambda(q,\psi)}$ be survival probability and $s_q = \partial s/\partial q$ its deployment derivative. The planner gain G_P^E adds the effect on survival to the private gain G_D^E:

$$G_P^E = G_D^E + \mathbb{E}[s_q(q,\psi)\{U(H(q,\psi)) - U(\underline{C})\}] = 0, \tag{57}$$

where

$$s_q(q,\psi) = -\lambda(\psi)\Delta\, s(q,\psi) < 0. \tag{58}$$

If surviving consumption exceeds the reservation floor, this additional term is negative.

D Fixed-date corrective charge

A charge on marginal conversion can implement the planner's allocation by pricing the additional disaster hazard, as in Acemoglu and Lensman (2024). Let G_D and G_P denote the private and planner marginal gains, and let $Z = \mathbb{E}[C_T^{-\gamma}]$. A charge τ paid in terminal consumption and rebated lump-sum changes the firm's condition to $G_D - Z\tau = 0$. At an interior planner allocation,

$$\tau^* = \frac{G_D - G_P}{Z}. \tag{59}$$

For economy-wide survivable damage, the wedge is the hazard term in Proposition 2. For adopter-only damage, it is the negative expected utility change from an additional event, weighted by the marginal hazard. Under extinction, the wedge is $-\mathbb{E}[s_q\{U(H) - U(\underline{C})\}]$ and the denominator is $\mathbb{E}[sH^{-\gamma}]$, since a financial charge can be collected only upon survival. Charges are in units of normalized old-technology terminal payoff per marginal conversion. At a boundary optimum, the corresponding inequality replaces the interior condition. The extinction charge depends on the independently specified reservation utility; it does not change private CRRA prices at a fixed allocation.

E Dynamic aggregation, valuation, and computation

E.1 Beliefs and dividend accounting

Let m_t and v_t be the posterior mean and variance based on productivity observations alone. The signal $dy_t = \psi\, dt + \sigma_N dZ_t$ gives

$$v_{t+h}^{-1} = v_t^{-1} + h/\sigma_N^2, \qquad m_{t+h} = v_{t+h}(m_t/v_t + \Delta y_t/\sigma_N^2). \tag{60}$$

With cumulative disaster count N_t and exposure $E_t = \int_0^t q_s ds$, the full posterior is

$$\pi_t(\psi) = \frac{\varphi_N(\psi; m_t, v_t)\lambda(\psi)^{N_t} e^{-E_t\lambda(\psi)}}{\int \varphi_N(z; m_t, v_t)\lambda(z)^{N_t} e^{-E_t\lambda(z)} dz}. \tag{61}$$

Here $\varphi_N(\cdot; m, v)$ is the normal density. The full posterior mean $\widehat{\psi}_t$ generally differs from m_t. Extinction decisions occur only before the event, when $N_t = 0$. The posterior is not approximated as Gaussian.

For case j, the displayed productivity-news path has $m_0 = 0$, $v_0 = 0.014204^2$, and

$$m_t^{(j)} = m_4^{(j)} \left(\frac{v_0 - v_{\min(t,4)}}{v_0 - v_4}\right)^{1/2}. \tag{62}$$

Case	Signal-only mean at year four
Economy-wide survivable loss	0.01599
Adopter-only loss	0.02521
Extinction: end of consumption	0.01198
Extinction: fixed reservation	0.02197

The endpoint $m_4^{(j)}$ is selected separately to match market adoption of 56.13% at year four. The same endpoint and observations are then used for the no-disaster and planner comparisons within that case. The common square-root path shape concentrates news earlier in the learning process and permits all four cases to reach the target. We recover signal increments by inverting (60). This specifies a sequence of observations, not a sequence of true capability draws or an average over selected simulations. Survival is then applied separately under each deployment path.

Let R_O be old-dividend growth before disaster damage and let n count events in the next interval, and write $L_n = \prod_{j=1}^{n} \Omega_j$ for the retained fraction after those events, with $L_0 = 1$. Each $\Omega_j = 0.9e^{-Z_j}$, where the Z_j are independent exponential draws with rate 6.3, independent of capability. Following conversion to r, installed dividends are $a^+ = a + K(r) - K(q)$. The next states are

$$B' = BR_O L_n, \quad a' = a^+ e^{\Delta y}, \qquad \text{economy-wide losses,} \tag{63}$$

$$B' = BR_O, \quad a' = a^+ e^{\Delta y} L_n, \qquad \text{adopter-only losses.} \tag{64}$$

Consumption is $B'(1 - r + a')$; exposure rises by rh and counts by n. Installed productivity is recorded by $x = \log[a/K(q)]$, set to zero before adoption. Economy-wide losses scale all dividends together; adopter-only losses change their composition. Size integration uses $\mathbb{E}[\Omega^b] = 0.9^b\, 6.3/(6.3 + b)$ for $b > -6.3$. Economy-wide-loss moments are analytic; adopter-only quadrature matches the relevant negative moments. Extinction ends all financial payoffs.

E.2 Decisions and timing

At a decision date, current consumption is paid from the installed firms. Firms then choose conversion, which reduces the scale of subsequent dividend flows. The next flow arrives after productivity news and disaster realization. The recursion uses hC_t^ρ for the current flow and $V_T = h^{1/\rho} C_T$ for the final flow. There is no terminal capital liquidation. Thus the conversion cost is the permanent dividend-scale loss, not an additional charge imposed on top of that loss.

Let $\mathcal{A}_t$ denote an adopter's value per unit of its current dividend, and $\mathcal{W}_t(u)$ an unconverted firm's value including its option to adopt. For a proposed aggregate action r, compare

$$J_t(u; r) = [1 - \kappa_\epsilon(u)]\mathbb{E}_t[M_{t,t+h} R_A\{h + \mathcal{A}_{t+h}\}], \tag{65}$$

$$L_t(u; r) = \mathbb{E}_t[M_{t,t+h} R_O\{h + \mathcal{W}_{t+h}(u)\}]. \tag{66}$$

The dividend growth factors include incidence-specific losses; the financial terms are zero on extinct branches. The pricing kernel is (32). An atomistic firm takes aggregate deployment and hazard as given. Aggregate consistency requires that firms choosing immediate conversion generate the proposed r. The solver uses cost ordering and a marginal crossing, allowing corner solutions; if there are multiple downward crossings it selects the first. This is a numerical selection rule, not a proof of equilibrium uniqueness.

The planner instead chooses $r \in [q, 1]$ to maximize the recursive value using the same transition probabilities, dividend losses, and extinction convention. This includes deployment's effect on

survival information. Under a fixed reservation flow, the death-branch continuation at date $t+h$ is

$$D_{t+h} = \underline{c} C_0 \left[h \sum_{j=0}^{(T-t-h)/h} \beta^j \right]^{1/\rho}. \tag{67}$$

This expression is independent of the pre-extinction consumption level. Because the fallback is fixed in initial units, old-dividend scale is an additional state; it cannot be removed by homogeneity. Its numerical coordinate is $b_t = \log(B_t/B_0) - gt$. The planner and private valuation use the same fixed-initial-units convention.

E.3 Hazard, risk reporting, and approximation

The full-deployment hazard is

$$\lambda(\psi) = \ell_{\max} \left[1 + \exp\left\{ -\alpha_D - b_D \frac{\psi - \psi_\star}{\bar{s}} \right\} \right]^{-1}. \tag{68}$$

Here $\ell_{\max} = 0.225$, $\psi_\star = 0.01056136$, and $\bar{s} = 0.01107202$ is a fixed scale, not posterior uncertainty. Define

$$L_0 = -\log(1 - 0.1033336890)/22, \qquad L_1 = -\log(1 - 0.2354116359)/22.$$

Then $\alpha_D = \log[L_0/(0.10 - L_0)]$ and $b_D = \log[L_1/(0.10 - L_1)] - \alpha_D$. The hazard scale is 2.25, giving an annual intensity cap of 0.225. All numerical exercises use this schedule. Appendix G describes the fixed-date terminal-payoff construction.

Reported perceived annual risk is

$$p_t^{(1)} = 1 - \int e^{-q_t \lambda(\psi)} \pi_t(\psi) d\psi. \tag{69}$$

It is a one-year, frozen-deployment probability, not the latent true hazard or a forecast integrating future adoption. The no-disaster adoption comparison applies this formula and survival updating to the no-disaster deployment path, without resolving that path under disaster risk.

Solutions use quarterly decisions, a signal grid with 43 nodes, 26 installed-productivity nodes (37 for random adopter-only losses), and interpolation over cumulative exposure. The economy-wide-loss, adopter-only, and end-of-consumption displays use adoption grid spacing 0.05. The fixed-reservation case uses 0.10 spacing and seven old-dividend-scale nodes, with quadrature over old-economy shocks. Transition probability mass, marginal-indifference residuals, and cost ordering are checked. These are exploratory finite-grid solutions; the comparisons do not constitute a joint convergence test across every state dimension or a global uniqueness proof.

The displayed yield line adjusts only the quarterly conversion-timing component. Specifically, it smooths the nonnegative sequence $-\log(C^+/C)/(\mathrm{IES}\, h)$ by penalizing second differences while preserving its sum, and substitutes that sequence into the raw yield. All other pricing components and the adoption policy remain unchanged. This visual approximation is not an alternative equilibrium or a correction to the pricing kernel.

E.4 Dynamic option-tail calculation

The option experiment holds the date fixed at year four and matches inherited deployment to $q = 0.5613$ separately in each economy using the same news-path shape as the time-series illus-

trations. The innovation raises the signal-only posterior mean by one monthly standard deviation, approximately 0.002897, and changes the observed productivity signal consistently with the Gaussian filter. Subsequent adoption and arrival/survival learning follow the solved equilibrium policies. The fixed-reservation consumption flow is $0.60C_0$. Survivable loss sizes and the aggregate equilibrium are the same as in the time-series illustrations.

Let P^j_{t+1} be the ex-dividend stock price one year ahead and $M_{t,t+1}$ the product of quarterly financial pricing kernels. Define the normalized financial pricing measure by weights proportional to $M_{t,t+1}$ and let $F^j_t = \mathbb{E}^Q_t[P^j_{t+1}]$. Under extinction, all financial payoffs and financial pricing weights are zero after termination; the normalized measure therefore conditions on financial survival. For $a = 0.10$, the plotted cumulative tail probabilities are

$$p^-_{j,t} = Q_t\Big(P^j_{t+1} \le (1-a)F^j_t\Big), \tag{70}$$

$$p^+_{j,t} = Q_t\Big(P^j_{t+1} \ge (1+a)F^j_t\Big). \tag{71}$$

These probabilities include all outcomes beyond the respective thresholds. The figure reports the after-news minus before-news difference, multiplied by 100. Each forward and pricing measure is recomputed after the innovation. The empirical specification approximates cumulative tail probabilities using finite option spreads farther from the forward, at $a = 0.20$ with strike-band width 0.10.

The market claim pays aggregate dividends. The adopted claim receives frontier growth and adopter losses. The non-adopter is the highest-cost firm, with conversion cost $\kappa + \epsilon$. It pays old-technology dividends and any economy-wide survivable loss until adoption, and retains the option to convert under the equilibrium policy. Its value is computed by backward pricing under the same aggregate policy and recursive utility as the other claims. Two independently scrambled Sobol simulations use 1,048,576 paths each, with common draws before and after the news. Pricing checks include dividends and terminal stock values; option payoffs use ex-dividend stock prices.

F Extinction inside and outside the risk aggregator

F.1 CRRA prices and the value of continued existence

In the fixed-date model, write $u(H) = H^{1-\gamma}/(1-\gamma)$ for utility from surviving consumption, U_E for the utility assigned to extinction, and $s(q,\psi)$ for survival probability. Welfare is

$$\mathcal{W}(q) = \mathbb{E}[s(q,\psi)u(H(q,\psi)) + (1 - s(q,\psi))U_E]. \tag{72}$$

Its marginal change is

$$\mathcal{W}_q = \underbrace{\mathbb{E}[sH^{-\gamma}H_q]}_{\text{private valuation}} + \underbrace{\mathbb{E}[s_q\{u(H) - U_E\}]}_{\text{value of survival lost}}. \tag{73}$$

A lower U_E makes the second term more negative wherever $s_q < 0$, while leaving the first term unchanged. Risk aversion specifies the curvature of utility over consumption. It does not fix the utility difference between being alive and being extinct. Equivalently, subtracting U_E from both outcomes gives zero utility in extinction and an intercept $-U_E$ in surviving utility. That intercept does not affect marginal consumption utility but matters when choices alter survival.

For $\gamma > 1$, the specification $U_E = u(\underline{C})$ permits an arbitrarily severe welfare penalty as $\underline{C} \downarrow 0$, without changing private CRRA prices at a fixed allocation. Under regularity conditions this can make optimal deployment arbitrarily small. With linear deployment exposure, the marginal

hazard is positive at $q = 0$, so a finite penalty can make zero deployment optimal. Conversely, a high reservation utility makes extinction less costly to the planner. These statements concern an independent welfare parameter, not a parameter that changes the financial recovery in extinction.

The same distinction holds with additive utility paid over time. Assigning zero utility after death is incomplete without specifying the intercept of utility while alive. In particular, the unshifted formula $C^{1-\gamma}/(1-\gamma)$ is negative for $\gamma > 1$; deleting its future terms would make a shorter life attractive. A sensible welfare comparison requires an explicit value-of-life normalization. This does not invalidate CRRA pricing of surviving claims, which depends on marginal utility rather than that intercept.

F.2 A binomial continuation problem

Use the notation of Section 3: $\rho = 1 - \eta$, η is inverse IES, γ is risk aversion, and s is next-period survival probability. To make the distinction transparent, suppose the surviving continuation value is the known positive number A and the extinction continuation value is $D > 0$. The continuation terms in the two recursions are

$$\text{end of consumption:} \quad sA^{\rho}, \tag{74}$$

$$\text{reservation utility:} \quad [sA^{1-\gamma} + (1-s)D^{1-\gamma}]^{\rho/(1-\gamma)}. \tag{75}$$

Both are multiplied by β and added to current utility hC^{ρ}. The first places survival outside the conditional risk certainty equivalent; the second treats extinction as an outcome inside it. For $\gamma > 1$, $D \downarrow 0$ makes $D^{1-\gamma}$ diverge. Zero consumption in an outcome that remains inside the risk aggregator is therefore not the same operation as ending the consumption problem.

For fixed s, A, and surviving consumption growth, let

$$Q^{\text{res}} = [sA^{1-\gamma} + (1-s)D^{1-\gamma}]^{1/(1-\gamma)}.$$

The ratio of the prices of the same next-period surviving payoff is

$$\frac{P^{\text{res}}}{P^{\text{end}}} = \left(\frac{Q^{\text{res}}}{A}\right)^{\gamma-\eta}. \tag{76}$$

When $D < A$ and $\gamma > \eta$, this ratio is below one and decreases as D falls. It is a level effect on all surviving prices at the current node. It does not determine their relative state-price weights at an earlier node where news about s arrives. Those weights depend on how news changes continuation values. A stronger response to hazard news can coexist with lower financial prices.

The direct short-yield effect follows from the same comparison. Write $s = e^{-\lambda h}$ and $z = (A/D)^{\gamma-1}$, holding surviving prospects and the reservation value fixed. Then

$$\frac{Q^{\text{res}}}{A} = [s + (1-s)z]^{-1/(\gamma-1)}, \qquad y^{\text{res}} - y^{\text{end}} = \frac{\gamma-\eta}{(\gamma-1)h}\log[s + (1-s)z]. \tag{77}$$

The end-of-consumption bond price is survival probability times the conditional price, so its direct yield adjustment is λ. Expanding the log term as $h \downarrow 0$ gives

$$y^{\text{res}} - y^{\text{end}} = \lambda\frac{\gamma-\eta}{\gamma-1}(z-1) + O(h), \tag{78}$$

which yields (35). At a fixed date, a higher surviving value A also raises z when D is fixed. The

discounting effect therefore strengthens as the surviving economy becomes more valuable.

F.3 An imputed reservation value for ending consumption

Allow the surviving continuation value to be uncertain again and let A_t be its certainty equivalent conditional on survival. We can match the continuation *level* of the end-of-consumption recursion by choosing an extinction value D^*_{t+h} satisfying

$$\left[s_t(A_t)^{1-\gamma} + (1-s_t)(D^*_{t+h})^{1-\gamma}\right]^{\rho/(1-\gamma)} = s_t(A_t)^{\rho}. \tag{79}$$

For $0 < s_t < 1$, $\rho > 0$, and $\gamma > 1$, solving gives

$$\boxed{D^*_{t+h} = A_t \left[\frac{s_t^{(1-\gamma)/\rho} - s_t}{1-s_t}\right]^{1/(1-\gamma)}}. \tag{80}$$

This value scales with surviving continuation utility. It is not a fixed consumption floor, and it need not represent any literal consumption or bequest after death. It is a way to express the preference over uncertain lifetimes implicit in (27) using the reservation-value algebra.

At the parameters used in Section 3, $\gamma = 4$ and $\rho = 1/2$, the ratio is

$$\frac{D^*_{t+h}}{A_t} = \left(\frac{s_t^{-6} - s_t}{1-s_t}\right)^{-1/3} = \left(1 + s_t^{-1} + \cdots + s_t^{-6}\right)^{-1/3}. \tag{81}$$

For a short interval with survival close to one,

$$\frac{D^*_{t+h}}{A_t} \longrightarrow \left[1 + \frac{\gamma - 1}{\rho}\right]^{-1/(\gamma-1)} = 7^{-1/3} \simeq 0.523. \tag{82}$$

The 52.3% figure is a fraction of the *surviving lifetime certainty equivalent*, not of current consumption. Future adoption and future disaster risk affect that certainty equivalent, so they also affect the imputed value.

This is an equivalence of preference functionals only if the rule in (80) applies to counterfactual choices as well as to the reported equilibrium. For example, differentiating the matched continuation term with respect to the surviving certainty equivalent gives $s_t\rho(A_t)^{\rho-1}$. To obtain the same derivative from the reservation expression, one must include the derivative of D^*_{t+h} with respect to A_t. A planner changing survival must also include its dependence on s_t. Computing D^* along one allocation and then freezing it while repricing assets or changing policy matches utility levels but generally fails to match prices or choices. In particular, (32) for a fixed nonfinancial fallback cannot be reused without these additional derivatives.

F.4 News about a later extinction event

A finite-horizon example separates the response to news from the discounting of surviving payoffs. Dates are $0, 1, \ldots, N$. Consumption is one at dates zero and one. At date one, investors learn a probability p of extinction before date two. If they survive, consumption remains one through date N and there is no further extinction risk. In the reservation case, extinction instead delivers a nonfinancial consumption flow $d > 0$ from date two through N. All bonds pay zero after extinction in both cases.

For unit decision intervals, define the remaining annuity weight $L_N = \sum_{j=0}^{N-2} \beta^j$ and $Z(p) = 1 - p + pd^{1-\gamma}$. The two date-one continuation values satisfy

$$(V_1^{\text{end}})^\rho = 1 + \beta L_N (1-p), \tag{83}$$

$$(V_1^{\text{res}})^\rho = 1 + \beta L_N Z(p)^{\rho/(1-\gamma)}. \tag{84}$$

If the news at date one can be either safe or dangerous, the relative state-price tilt at date zero is

$$\frac{M_{01}(\text{dangerous})}{M_{01}(\text{safe})} = \left[\frac{V_1(\text{dangerous})}{V_1(\text{safe})}\right]^{\eta-\gamma}. \tag{85}$$

The common date-zero certainty equivalent cancels from this ratio. Meanwhile, a date-one bond promising one at date N has price

$$B_{1N}^{\text{end}} = \beta^{N-1}(1-p), \tag{86}$$

$$B_{1N}^{\text{res}} = \beta^{N-1}(1-p)Z(p)^{(\eta-\gamma)/(\gamma-1)}. \tag{87}$$

Only one extinction event is possible, regardless of N.

With $\gamma = 4$, IES 2, $\beta = e^{-0.03}$, $d = 0.4$, and equally likely news $p = 0$ or $p = 0.01$, the $N = 30$ state-price tilt is 6.9% under end of consumption and 16.3% under reservation consumption. Yet the date-zero bond price is approximately 99.5% of its no-extinction value in the first case and 91.7% in the second. Thus the reservation specification can have the *stronger* hedge against hazard news and the *larger* reduction in bond prices. A claim exchanging current consumption for a stream of future productivity gains need not become more attractive merely because the relative tilt increases.

G Risk-neutral density construction

Recall that ψ is quality, q is adopted mass, $H(q,\psi)$ is aggregate wealth before economy-wide disaster losses, and $\Delta = T - t^*$ is the remaining horizon. The loading $A_2 = A_2(\Delta)$ comes from the fixed-date productivity process with mean-reversion speed ϕ and volatility σ_O. The comparison restores the continuous common old-technology payoff that was normalized out of the adoption condition. Write this common factor as e^{ξ_O}, where ξ_O is its log payoff and v_O its variance, independently of ψ and the disaster process. Conditional on current old-technology productivity, the Gaussian loading in (47) gives

$$\xi_O \sim \mathcal{N}(0, v_O), \quad v_O = \frac{\sigma_O^2}{\phi^2}\left[\Delta - \frac{2(1-e^{-\phi\Delta})}{\phi} + \frac{1-e^{-2\phi\Delta}}{2\phi}\right]. \tag{88}$$

The mean is a scale normalization. This independent factor cancels from the adoption conditions but gives all three claims continuous payoff distributions, including the non-adopter. At the endpoints of the conversion-cost distribution, $\kappa_L = \kappa - \epsilon$ and $\kappa_H = \kappa + \epsilon$. For firm claim $i \in \{L, H\}$, write X_i for its terminal payoff and $\mathbf{1}\{\cdot\}$ for an indicator. With disaster count N and retained fraction ω, it pays

$$X_i = e^{\xi_O}\left[\mathbf{1}\{\text{adopt}_i\}(1-\kappa_i)e^{A_2\psi} + \mathbf{1}\{\text{do not adopt}_i\}\right]\omega^N, \qquad C_T = e^{\xi_O}H(q,\psi)\omega^N. \tag{89}$$

Let P be the physical posterior distribution of payoffs and Q its normalized marginal-utility-

weighted counterpart. With risk aversion γ, all claims share the same pricing measure:

$$\frac{dQ}{dP} = \frac{C_T^{-\gamma}}{\mathbb{E}^P[C_T^{-\gamma}]}. \tag{90}$$

Under Q, ξ_O is Gaussian with mean $-\gamma v_O$ and variance v_O. Writing $f_P(\psi)$ for the physical posterior density of quality, its marginal density under Q is proportional to

$$f_P(\psi)H(q,\psi)^{-\gamma}\exp\{\Lambda(q,\psi)(\omega^{-\gamma}-1)\}, \tag{91}$$

and $N \mid \psi$ is Poisson with mean $\Lambda(q,\psi)\omega^{-\gamma}$. The exponent differs from the adoption multiplier $\chi = \omega^{1-\gamma} - 1$ because a probability receives the marginal-utility weight $\omega^{-\gamma N}$, whereas the payoff gain in that condition supplies an additional ω^N. Applying a different tilt to each claim would be inconsistent with a common household stochastic discount factor.

The density illustration and Figure 1 use the same fixed-date parameters: $\gamma = 4$, $A_2 = 13.1935$, $\widehat{s} = 0.01346$, $\Delta = 16$, $\kappa = 0.10$, $\epsilon = 0.015$, and $\omega = 0.80$. Costs, the prior and signal parameters, and the annual hazard schedule agree with the dynamic calibration in Table 2. The hazard is the bounded logistic function in Appendix E, with cap 0.225 and linear deployment exposure. Conditional on deployment q, the Poisson mean is $q\lambda(\psi)\Delta$. The fixed 20% loss isolates the static mechanism; the dynamic model allows random severity. Holding this hazard schedule fixed, we solve for the belief that produces private deployment $q = 0.50$. The higher belief is 0.01000; the lower belief, -0.00119, lies half a posterior standard deviation below the low-cost firm's entry threshold and produces $q = 0$.

H Empirical data construction

Option probabilities are constant-maturity, strike-band digital approximations constructed from OptionMetrics and a Databento extension. The main-text equations state the put and call spreads, discounting, and forward-relative thresholds. For QQQ and IWM, each wing's Databento probability is adjusted by subtracting the median Databento-minus-OptionMetrics level difference on overlapping dates. OptionMetrics observations are retained through their final date; adjusted Databento observations are appended afterward. The adjustment is applied separately to downside and upside probabilities. SPY-labelled probabilities use the saved SPX series splice and are paired with SPY returns.

Early-2022 OptionMetrics observations extend the usable probability changes back to February 2022, permitting the first full 13-month regression window to end in February 2023. Saved overlap checks reproduce existing probabilities and monthly observations. Available expirations are interpolated to a one-year maturity. Month-end probabilities use the last available observation separately for each wing; monthly returns compound daily returns. Probability changes and returns are in percentage points. The final complete month is August 2026. Every regression includes an intercept and QQQ and SPY returns and uses Newey–West covariance estimates with three lags.

A vendor change can affect probability changes near the join even after a level adjustment. ETF American exercise, forward estimation, strike interpolation, and finite spread widths are additional approximations. These objects average digital probabilities over strike bands rather than measure exact pointwise tails. Ramp's monthly adoption level is displayed only over its available dates on a separate axis.

The portfolio comparison matches December 31, 2025 CRSP Mutual Fund Holdings to the total workforce-task exposure scores of Eisfeldt et al. (2023), using date-valid security identifiers and

CRSP–Compustat links. The October 2025 score release reflects March 2022 workforces and March 2023 task assessments. We exclude non-equity positions and combine share classes. Reconstructed November 2022 NYSE exposure quintiles define H as the bottom quintile and A as the top; these approximate the authors' sorts rather than reproduce exact portfolio memberships. The main-text shares count each firm once and condition on membership in A or H, excluding the middle quintiles and unclassified firms. IWM has 232 A and 162 H firms, giving a denominator of 394; QQQ has 26 A and 9 H firms, giving a denominator of 35. These extremes represent 37.3% of IWM's 1,055 classifiable firms and 41.7% of QQQ's 84. Scores cover 54.4% of IWM's 1,941 firms and 84.0% of QQQ's 100 firms. Unclassified firms are excluded, not assigned to H. The comparison is therefore conditional on coverage and describes labor-task exposure, not firm size or other determinants of adoption.